

ECHO: EXPLICITLY COUPLED HIERARCHICAL OPTIMIZATION FOR REASONING AND TOOL USE

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ABSTRACT

Tool-using language-model agents couple reasoning to tool learning: reasoning determines the trajectories used to adapt the tool policy, whose response in turn changes task performance. We introduce ECHO, *Explicitly Coupled Hierarchical Optimization*, which optimizes reasoning for the adapted tool behavior induced by its commitment. Each round runs K standardized, clipped group relative policy optimization (GRPO) tool updates from a common base, evaluates fresh task roll-outs, and propagates return through adaptation. Our theory gives three guarantees. Exact response-aware updates make the reasoning parameters and induced tool response converge linearly to the Stackelberg equilibrium. K exact-gradient tool updates before each reasoning update make both approach that equilibrium up to bounded finite-adaptation errors, which vanish in the large- K limit. For grouped stochastic updates, a leave-one-out proxy applies relative trajectory advantages in all K tool updates and the reasoning update. Under conditional bias and variance bounds, the reasoning parameters approach their value at the Stackelberg equilibrium within a bounded expected squared error. At fixed K , increasing the query count at every tool update and task evaluation recovers the deterministic K -step gradient. We then empirically validate the GRPO implementation of ECHO on arithmetic and knowledge-intensive reasoning benchmarks. The results show that its learned policies approach a Stackelberg equilibrium and that ECHO outperforms state-of-the-art tool-calling algorithms across the evaluated tasks.

1 INTRODUCTION

Language-model agents equipped with external tools work through a query by interleaving reasoning with tool calls for retrieval, search, or code execution. A reasoning policy identifies what information or computation to seek. A tool policy selects the action, and the observation shapes later decisions. Tool actions have local validity and usefulness scores, while final task performance evaluates the complete trajectory. The two policies therefore form one adaptive system.

The training loop makes this coupling central to what both policies learn: each policy shapes the data and feedback used to train the other. A change in the reasoning policy changes the histories and tool requests used to adapt the tool policy. Once adapted to those requests, the tool policy changes the observations available to the reasoning policy, its later decisions, and ultimately the task return. We call this causal dependence of post-adaptation tool behavior on the reasoning decision *decision-dependent non-stationarity*. A reasoning update must therefore be evaluated with the tool behavior it induces through learning. How can reasoning account for that learned response?

Our solution is to train the reasoning policy accounting for the performance of the adapted reasoning-tool pair that it induces. After the reasoning policy commits to its current parameters, the tool policy adapts for K steps from a common base. Fresh queries are used to evaluate the adapted pair. We propagate the resulting task return through the adaptation path. This Stackelberg leader-response sequence defines *Explicitly Coupled Hierarchical Optimization* (ECHO), whose reasoning update combines the direct effect with the learned tool response.

Each ECHO round begins with K GRPO updates to the tool policy from a common base. Fresh query groups compare tool trajectories and assign greater credit to those with higher relative returns. Fresh task queries then evaluate the adapted reasoning-tool pair. Differentiating their return through all K updates combines the reasoning policy’s direct effect with its indirect effect through the tool

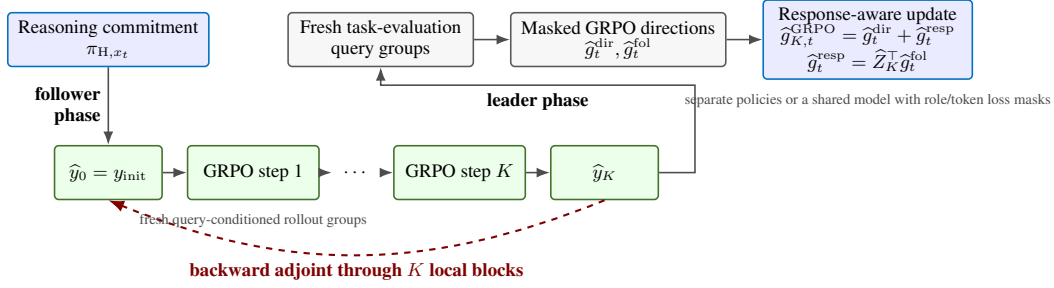


Figure 1: ECHO adapts the tool policy for K GRPO steps under a fixed reasoning commitment, evaluates fresh task rollouts, and propagates their return backward through every tool update.

policy it induces. This inner–outer structure closely resembles meta-learning: tool adaptation forms the inner loop, and reasoning optimization forms the outer loop. The roles may use separate policies or a shared model with role-specific masks.

We make the following main contributions. We formulate the Stackelberg learning objective and derive the reasoning leader’s direct-plus-response gradient. We develop a two-phase GRPO algorithm with a common tool-policy base, fresh query groups in each phase, and task return propagated through all K tool updates. Our analysis establishes a hierarchy of three results spanning the successive approximations used in the algorithm. (i) With the exact learned tool response, exact response-aware updates make the reasoning parameters and induced tool response converge linearly to the Stackelberg equilibrium. (ii) With K exact-gradient tool updates before each reasoning update, the reasoning parameters and tool response approach that equilibrium up to bounded finite-adaptation errors, which vanish in the large- K limit. (iii) For grouped stochastic updates, we analyze a tractable proxy based on REINFORCE leave-one-out (RLOO) that retains the query-group structure of GRPO. Under the stated bias and variance bounds, K sampled tool updates followed by a sampled reasoning update, using group-relative advantages throughout, make the reasoning parameters approach their equilibrium value within a bounded long-run expected squared error. At fixed K , increasing query counts at every tool update and task evaluation makes the sampled gradient approach the corresponding deterministic K -step gradient. Together, these results suggest that ECHO is likely to approach the Stackelberg equilibrium with bounded error in practice.

2 RELATED WORK

Learning reasoning and tool use. Tool-augmented language models combine generation with external action through prompting, self-supervised API-use training, instruction tuning over tool libraries, and reinforcement learning of integrated reasoning–tool trajectories (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024; Li et al., 2025). These methods establish joint reasoning–action trajectories and increasingly train tool selection itself. ECHO studies the learning feedback within such a trajectory. Reasoning determines the histories and requests used to adapt the tool policy, while the adapted behavior changes subsequent observations and final task return.

Differentiating learned responses. Meta-learning and bilevel optimization differentiate an outer objective through an inner learning path or solution (Finn et al., 2017; Franceschi et al., 2018; Maclaurin et al., 2015). Analyses of truncated and stochastic hypergradients quantify how finite or sampled inner procedures approximate this response (Shaban et al., 2019; Grazi et al., 2020; Ji et al., 2021; Grazi et al., 2021), while MAML convergence theory studies the resulting outer dynamics (Fallah et al., 2020). Implicit Stackelberg learning interprets the adapted component as a follower response (Fiez et al., 2020). Recent language-model systems realize this response in several ways. STA-RLHF and Gauthier et al. account for how a policy changes the preference or reward model trained on its outputs (Makar-Limanov et al., 2024; Gauthier et al., 2026). SLHF conditions a follower refinement on a leader-generated candidate, whereas Bi-NAC optimizes a feedback-generating critic through the adaptation of a reasoning actor (Pásztor et al., 2026; Singh et al., 2026). ECHO places the response in the tool policy, which adapts for K steps under a reasoning commitment on the same trajectory. Its analysis connects exact, finite-step, and sampled responses

to convergence and residual error at the Stackelberg equilibrium. For the finite-step result, ECHO combines response error with a standard projected-gradient contraction (Nesterov, 2004) to bound the resulting distance from equilibrium.

Grouped gradients through adaptation. GRPO and leave-one-out REINFORCE both draw several rollouts per query and assign credit relative to the group. GRPO standardizes group returns and clips tokenwise likelihood ratios (Shao et al., 2024), while the leave-one-out construction forms each rollout’s baseline from the other returns in its query group (Kool et al., 2019; Ahmadian et al., 2024). ECHO uses GRPO for practical training and the latter construction as a tractable grouped proxy in Section 6. Differentiating through its policy-gradient updates draws on higher-order stochastic-computation-graph and meta-gradient methods (Foerster et al., 2018; Mao et al., 2019; van Krieken et al., 2021; Tang et al., 2021). ECHO specializes them to the first- and second-order blocks linking reasoning to K tool updates. Appendix B develops these connections in greater detail.

3 HIERARCHICAL RESPONSE-AWARE OBJECTIVE

Let $x \in X \subset \mathbb{R}^{d_x}$ and $y \in Y \subset \mathbb{R}^{d_y}$ be the trainable coordinates of the reasoning (leader) policy $\pi_{H,x}$ and tool (follower) policy $\pi_{L,y}$. Both utilities depend on their complete joint trajectory. Algorithm 1 realizes these coordinates with separate policies or a shared model with role-specific loss masks. Section 6 uses role-isolated coordinates for its stochastic analysis.

Queries, trajectories, and utilities. For $q \sim P_Q$, a complete episode is $\tau = (m_h, a_h, o_h)_{h=1}^{T_{\text{ep}}}$. With c_{h-1} denoting the preceding history and $\mathcal{A}_{\text{tool}}(q)$ the available tools,

$$p_{x,y}(\tau \mid q, \mathcal{A}_{\text{tool}}(q)) = \prod_{h=1}^{T_{\text{ep}}} \pi_{H,x}(m_h \mid c_{h-1}) \pi_{L,y}(a_h \mid c_{h-1}, m_h) \times P_{\text{env}}(o_h \mid c_{h-1}, m_h, a_h), \quad (1)$$

where the parameter-independent execution kernel is a point mass for a deterministic tool. Let $\mathcal{R}_L = r_{\text{valid}}$ and $\mathcal{R}_H = r_{\text{task}}$ denote the realized tool-validity and final task returns. Either may include parameter-independent shaping. Let $\mathcal{H}_{\text{tool}}$ be expected tool-policy entropy and $\mathcal{K}_L, \mathcal{K}_H$ expected forward KL penalties to fixed role-specific references. For nonnegative coefficients $\lambda = (\lambda_{\text{ent}}, \lambda_{\text{valid}}, \lambda_L, \lambda_H)$, define

$$u_L(x, y) = \lambda_{\text{ent}} \mathcal{H}_{\text{tool}}(x, y) + \lambda_{\text{valid}} \mathbb{E}_{q,\tau}[\mathcal{R}_L(q, \tau)] - \lambda_L \mathcal{K}_L(x, y), \quad (2)$$

$$u_H(x, y) = \mathbb{E}_{q,\tau}[\mathcal{R}_H(q, \tau)] - \lambda_H \mathcal{K}_H(x, y). \quad (3)$$

The expectation includes a fresh query and all randomness in Eq. (1). Appendix C.1 records the entropy–KL accounting. Section 6 specifies the utility-level return scaling and complete regularizer derivatives used by the analyzed estimators.

3.1 EXACT RESPONSE-AWARE GRADIENT

For a reasoning commitment x , define

$$\xi(x) := \arg \max_{y \in Y} u_L(x, y), \quad F(x) := u_H(x, \xi(x)). \quad (4)$$

Under the conditions in Section 5, $\xi(x)$ is unique and interior, and F has a unique maximizer x^* . Define $y^* := \xi(x^*)$. The pair (x^*, y^*) is called the *Stackelberg equilibrium*. For vector derivatives, our Jacobian and mixed-block convention is

$$\nabla_{yx}^2 u_L := D_x(\nabla_y u_L) \in \mathbb{R}^{d_y \times d_x}, \quad g^{\text{dir}} := \nabla_x u_H, \quad g^{\text{fol}} := \nabla_y u_H.$$

Implicit differentiation and the chain rule give

$$D\xi(x) = -[\nabla_{yy}^2 u_L(x, \xi(x))]^{-1} \nabla_{yx}^2 u_L(x, \xi(x)), \\ g_{\text{ex}}(x) := \nabla F(x) = g^{\text{dir}}(x, \xi(x)) + D\xi(x)^\top g^{\text{fol}}(x, \xi(x)). \quad (5)$$

The second term carries the learned tool response and corrects for decision-dependent non-stationarity.

3.2 FINITE RESPONSE-AWARE UPDATE

Every response begins from the x -independent common base y_{init} :

$$y^{(0)}(x) = y_{\text{init}}, \quad y^{(s+1)}(x) = y^{(s)}(x) + \eta_L \nabla_y u_L(x, y^{(s)}(x)), \quad \xi_K(x) := y^{(K)}(x). \quad (6)$$

For $Z_s := D_x y^{(s)}(x)$, define $H_{yc}^{(s)} := \nabla_{yc}^2 u_L(x, y^{(s)}(x))$ for $c \in \{x, y\}$. Then

$$Z_{s+1} = (I + \eta_L H_{yy}^{(s)})Z_s + \eta_L H_{yx}^{(s)}, \quad Z_0 = 0.$$

Thus the finite-response objective, total derivative, and leader update are

$$\begin{aligned} F_K(x) &:= u_H(x, \xi_K(x)), \\ g_K(x) &:= \nabla F_K(x) = g^{\text{dir}}(x, \xi_K(x)) + Z_K(x)^\top g^{\text{fol}}(x, \xi_K(x)), \\ x_{t+1} &= \text{Proj}_X(x_t + \eta_H g_K(x_t)). \end{aligned} \quad (7)$$

The response is counted once, either by differentiating the composition or by adding the displayed term. Appendix C.2 derives the update. Appendices D–E quantify its finite- K error.

4 RESPONSE-AWARE GRPO FOR ECHO

ECHO runs GRPO in two phases. In the follower phase, each of K tool steps draws a fresh group per query, standardizes their returns, and takes one clipped tool-masked update. The leader phase uses an independent task batch to compute the masked direct and tool directions, then propagates the latter through all K steps of tool adaptation.

x and y denote the leader (reasoning) and follower (tool) policy parameters, respectively. With shared weights, we write $w(x, y) = w_{\text{core}} + P_H x + P_L y$ and use role/token masks to route the losses. Each round keeps x_t , resets y and its optimizer state, and discards the adapted follower after updating x . Appendices C.1 and C.3 describe the shared-model routing and reset.

Algorithm 1 Response-aware ECHO with K tool-policy GRPO adaptations

Require: Reasoning view π_{H,x_0} and common tool-policy base $\pi_{L,y_{\text{init}}}$, implemented separately or by a shared model with role/token masks. Step sizes η_L, η_H , horizon K , outer rounds N_{out} , query-batch sizes B_L, B_H , and rollout-group sizes $G_L, G_H \geq 2$.

- 1: **for** $t = 0, \dots, N_{\text{out}} - 1$ **do**
- 2: Commit to x_t and clone $\hat{y}_0 \leftarrow y_{\text{init}}$ with fresh optimizer state.
- 3: Set $\hat{Z}_0 \leftarrow 0$.
- 4: **for** $s = 0, \dots, K - 1$ **do** **▷ Follower phase: tool adaptation**
- 5: Draw the B_L entries of $\mathcal{Q}_{L,t}^{(s)}$ independently from P_Q .
- 6: For each batch b , hold $q_{L,t,b}^{(s)}$ fixed and generate G_L trajectories in $\mathcal{G}_{L,t,b}^{(s)}$ under $(x_t, \hat{y}_s)$.
- 7: Form $\hat{U}_{L,t,s}^{\text{GRPO}}$ and compute $\hat{g}_{L,s}^{\text{GRPO}}$.
- 8: Update $\hat{y}_{s+1}$ by Eq. (8). Retain the fixed-rollout record and complete query-group scores for the score-corrected reverse sweep, or propagate $\hat{Z}_{s+1}$ by Eq. (37).
- 9: **end for**
- 10: Draw the B_H queries in $\mathcal{Q}_{H,t}$ independently from P_Q . **▷ Leader phase: task evaluation and update**
- 11: For each batch b , hold $q_{H,t,b}$ fixed and generate G_H trajectories in $\mathcal{G}_{H,t,b}$ under $(x_t, \hat{y}_K)$.
- 12: Form $\hat{U}_{H,t}^{\text{reason}}$ and $\hat{U}_{H,t}^{\text{tool}}$ and compute $\hat{g}_t^{\text{dir}}$ and $\hat{g}_t^{\text{fol}}$.
- 13: Form $\hat{g}_{K,t}^{\text{GRPO}}$ using Eqs. (10)–(11) and update $x_{t+1} \leftarrow \text{Proj}_X(x_t + \eta_H \hat{g}_{K,t}^{\text{GRPO}})$.
- 14: Discard the adapted tool clone and its optimizer state.
- 15: **end for**
- 16: **return** $x_{N_{\text{out}}}$ and the response obtained by adapting y_{init} to it with the same K -step protocol.

The sampled follower step and fresh leader-phase directions are

$$\hat{g}_{L,s}^{\text{GRPO}} := \nabla_y \hat{U}_{L,t,s}^{\text{GRPO}}(x, y) \Big|_{(x_t, \hat{y}_s)}, \quad \hat{y}_{s+1} = \hat{y}_s + \eta_L \hat{g}_{L,s}^{\text{GRPO}}, \quad \hat{y}_0 = y_{\text{init}}, \quad (8)$$

$$\hat{g}_t^{\text{dir}} := \nabla_x \hat{U}_{H,t}^{\text{reason}}(x, y) \Big|_{(x_t, \hat{y}_K)}, \quad \hat{g}_t^{\text{fol}} := \nabla_y \hat{U}_{H,t}^{\text{tool}}(x, y) \Big|_{(x_t, \hat{y}_K)}. \quad (9)$$

For disjoint role coordinates, these masked task-return GRPO directions estimate the two role partials.

Let $\hat{H}_{yy,s}^{\text{GRPO}}$ and $\hat{H}_{yx,s}^{\text{GRPO}}$ be the score-corrected Jacobian blocks in Appendix C.4. Starting from $v_K = \hat{g}_t^{\text{fol}}$, the reverse sweep is

$$v_s = (I + \eta_L \hat{H}_{yy,s}^{\text{GRPO}})^\top v_{s+1}, \quad s = K-1, \dots, 1, \quad (10)$$

$$\hat{g}_t^{\text{resp}} = \eta_L \sum_{s=0}^{K-1} (\hat{H}_{yx,s}^{\text{GRPO}})^\top v_{s+1}, \quad \hat{g}_{K,t}^{\text{GRPO}} := \hat{g}_t^{\text{dir}} + \hat{g}_t^{\text{resp}}. \quad (11)$$

These products equal Appendix C.4’s forward recursion, so either route adds the same response term once. Section 6 retains this query-grouped structure in a tractable proxy and analyzes grouped trajectory estimates in both phases.

5 RESPONSE-AWARE CONVERGENCE THEORY

This section proves the first two analytical guarantees of ECHO. Exact response-aware updates make the reasoning parameters and induced tool response converge linearly to the Stackelberg equilibrium. With K exact-gradient tool updates before each reasoning update, both approach that equilibrium up to bounded finite-adaptation errors, which vanish in the large- K limit. We use expected utilities on compact trust regions $X \times Y$ and project the outer update.

5.1 EXACT RESPONSE-AWARE OPTIMIZATION

Assumption 1 (Smooth uniformly strongly concave follower). $X \subset \mathbb{R}^{d_x}$ and $Y \subset \mathbb{R}^{d_y}$ are nonempty, compact, convex, and full-dimensional, and both utilities are C^2 on an open neighborhood of $X \times Y$. For every x , $u_L(x, \cdot)$ is μ_L -strongly concave with L_L -Lipschitz gradient on Y , where $0 < \mu_L \leq L_L$. Its maximizer is interior, and $\sup_{X \times Y} \|\nabla_{yx}^2 u_L\| \leq M_{yx}$.

Assumption 2 (Strongly concave reduced objective). $F(x) = u_H(x, \xi(x))$ is μ_H -strongly concave with L_F -Lipschitz gradient on X , and its unique maximizer x^* is interior. Moreover,

$$0 < \eta_H < 2\mu_H/L_F^2, \quad \kappa_H := \sqrt{1 - 2\eta_H\mu_H + \eta_H^2 L_F^2} < 1.$$

Assumption 1 gives a unique C^1 response with $\|D\xi(x)\| \leq M_{yx}/\mu_L$. Appendix D gives the details.

Theorem 1 (Exact convergence with the response-aware gradient). *Under Assumptions 1 and 2, let $x_{t+1} = \text{Proj}_X(x_t + \eta_H g_{\text{ex}}(x_t))$ and $y_t = \xi(x_t)$. Then, for every $t \geq 0$,*

$$\|x_t - x^*\| \leq \kappa_H^t \|x_0 - x^*\|, \quad (12)$$

$$\|y_t - \xi(x^*)\| \leq (M_{yx}/\mu_L) \kappa_H^t \|x_0 - x^*\|, \quad (13)$$

$$F(x^*) - F(x_t) \leq (L_F/2) \kappa_H^{2t} \|x_0 - x^*\|^2. \quad (14)$$

Thus both parameter blocks converge linearly to the Stackelberg equilibrium (x^*, y^*) .

The response term also measures the displacement caused by freezing the adapted tool policy. Define

$$\bar{g}_H := \sup_{X \times Y} \|\nabla_y u_H(x, y)\| < \infty, \quad g^{\text{dir}}(x) := \nabla_x u_H(x, \xi(x)). \quad (15)$$

Proposition 1 (Response-aware and frozen-response stationary points). *Under Assumptions 1 and 2, if g^{dir} has an interior zero $x^\dagger$, then*

$$\|x^\dagger - x^*\| \leq \mu_H^{-1} \|D\xi(x^\dagger)^\top \nabla_y u_H(x^\dagger, \xi(x^\dagger))\| \leq \frac{M_{yx} \bar{g}_H}{\mu_L \mu_H}. \quad (16)$$

The first inequality is exact for the scalar quadratic family in Appendix G. Cancellation can nevertheless make the response term vanish in a coupled system.

The bound makes decision-dependent non-stationarity quantitative. At $x^\dagger$, the direct field is zero, so $\nabla F(x^\dagger)$ equals the response term. Strong concavity turns its norm into distance from x^* . The factors M_{yx}/μ_L and $\bar{g}_H$ measure response sensitivity and leader-utility variation along that response.

5.2 FINITE-STEP EXACT-GRADIENT TOOL ADAPTATION

Let $D_0 := \sup_{x \in X} \|y_{\text{init}} - \xi(x)\|$ and $\mathcal{T}_{D_0}(\xi(X)) := \{y : \inf_{z \in \xi(X)} \|y - z\| \leq D_0\}$.

Assumption 3 (Uniform finite-response regularity). $0 < \eta_L \leq 1/L_L$ and $\mathcal{T}_{D_0}(\xi(X)) \subseteq Y$. Uniformly on $X \times \mathcal{T}_{D_0}(\xi(X))$, the blocks $\nabla_{yy}^2 u_L$ and $\nabla_{yx}^2 u_L$ are Lipschitz in y with moduli ℓ_{yy}, ℓ_{yx} , while the leader partials satisfy

$$\|\nabla_x u_H(x, y) - \nabla_x u_H(x, y')\| \leq L_{Hx,y} \|y - y'\|, \quad (17)$$

$$\|\nabla_y u_H(x, y) - \nabla_y u_H(x, y')\| \leq L_{Hy,y} \|y - y'\|. \quad (18)$$

Put $\kappa_L := 1 - \eta_L \mu_L \in [0, 1)$. Contraction from y_{init} keeps every unprojected inner iterate in the stated tube, and Appendix D proves uniformly in x that

$$\|\xi_K - \xi\| \leq e_y(K), \quad \|Z_K - D\xi\| \leq e_Z(K), \quad \|g_K - g_{\text{ex}}\| \leq \varepsilon_K. \quad (19)$$

For fixed $0 < \kappa_L < 1$, $e_y(K) = O(\kappa_L^K)$ and $e_Z(K), \varepsilon_K = O(K\kappa_L^{K-1})$, so every error vanishes with K . Appendix D also covers $\kappa_L = 0$.

Theorem 2 (Bounded error after K exact-gradient tool updates). *Under Assumptions 1–3, fix $K \geq 1$ and let $x_{t+1} = \text{Proj}_X(x_t + \eta_H g_K(x_t))$ and $y_t = \xi_K(x_t)$, restarting every response from y_{init} . Then*

$$\|x_t - x^*\| \leq \kappa_H^t \|x_0 - x^*\| + \frac{\eta_H \varepsilon_K (1 - \kappa_H^t)}{1 - \kappa_H} =: D_{t,K}, \quad (20)$$

$$\|y_t - \xi(x^*)\| \leq e_y(K) + (M_{yx}/\mu_L) D_{t,K}, \quad (21)$$

$$\limsup_{t \rightarrow \infty} \|x_t - x^*\| \leq \eta_H \varepsilon_K / (1 - \kappa_H). \quad (22)$$

Thus K exact-gradient tool updates followed by the reasoning update give the explicit bounded errors above for both coordinates. The outer transient decays with t , and the remaining finite-adaptation errors vanish with K . Restarting from the common y_{init} makes $\xi_K(x)$ the same protocol for every reasoning commitment encountered by the algorithm.

Appendix E proves both theorems and the response-gap proposition, handles $\kappa_H = 0$, and records the associated utility bounds. Section 6 connects this deterministic finite- K guarantee to grouped trajectory estimates used in both the tool and reasoning updates.

6 A TRACTABLE MODEL OF THE GRPO RESPONSE

Algorithm 1 trains ECHO with standardized, clipped, tokenwise GRPO. Its response hypergradient must differentiate through K stochastic tool updates, while within-group standardization, clipping, and tokenwise likelihood ratios make the higher-order analysis difficult. We therefore seek an analytically tractable proxy that retains the central structure of GRPO: several rollouts answer each query, and each receives credit relative to the others in its group. REINFORCE leave-one-out (RLOO) (Kool et al., 2019; Ahmadian et al., 2024) provides such a proxy. For each query, it centers the raw return of a focal rollout by the mean return of the other rollouts. This baseline is conditionally independent of the focal trajectory, so pairing it with a full-trajectory likelihood ratio preserves the expected score gradient at the behavior point.

We apply RLOO throughout the proxy. We use fresh groups to compute relative trajectory advantages for all K tool updates and an independent batch of task query groups for the reasoning update. At each behavior point, the expectations of these local estimates recover each policy gradient and the second-order response blocks needed to differentiate through the sampled path. This yields the third guarantee, namely, under the stated bias and variance bounds, K grouped tool updates followed by a grouped reasoning update place the reasoning parameters within a bounded long-run expected squared error of their equilibrium value. At fixed K , increasing the query count at every tool update and task evaluation recovers g_K , the deterministic gradient through K exact-gradient tool updates.

Proposition 2 (First- and second-order RLOO derivative identities). *Under Assumption 4 (common support, dominated second derivatives, independent groups with conditionally iid rollouts, role isolation, and exact regularizers), for $i \in \{L, H\}$, $c, d \in \{x, y\}$, and any behavior point $\bar{\vartheta}$,*

$$\mathbb{E}_{\mathcal{D} \sim \mathbb{P}_{\bar{\vartheta}}} \left[\nabla_c \hat{\mathcal{L}}_i \Big|_{\vartheta = \bar{\vartheta}} \right] = \nabla_c U_i(\bar{\vartheta}), \quad (23)$$

$$\mathbb{E}_{\mathcal{D} \sim \mathbb{P}_{\bar{\vartheta}}} \left[D_d \nabla_c \hat{\mathcal{L}}_i \Big|_{\vartheta = \bar{\vartheta}} \right] = D_d \nabla_c U_i(\bar{\vartheta}). \quad (24)$$

These identities hold for every finite $G \geq 2$. The first estimates the leader and follower partials. The second retains the cross-role score product and estimates the yy and yx response blocks.

At tool step s , the yx block measures how the reasoning commitment changes that update, while the yy block carries earlier tool changes into later updates. Chaining these blocks across K steps constructs $\hat{Z}_K$, whose action on the fresh tool partial gives Eq. (11)’s response term. Appendix F.3 bounds the finite-group mean-square error by $O(1/N_q)$ across queries and $O(1/(N_q G))$ within a query for every $G \geq 2$.

Finite-batch error bound. Fresh batches generate each sampled tool state and sensitivity, and an independent task batch produces the reasoning update in Eq. (65). The term β_K measures its systematic shift from the deterministic K -step gradient, while Σ_K^2 is its conditional variance.

Theorem 3 (Bounded error with RLOO group-trajectory estimates). *Under Assumptions 1–3, fix K such that $\sup_x \|g_K(x) - \nabla F(x)\| \leq \varepsilon_K$ and impose Eq. (76). Then the projected iterates satisfy*

$$\begin{aligned} x_{t+1} &= \text{Proj}_X(x_t + \eta_H \hat{g}_{K,t}^{\text{RLOO}}), \\ \limsup_{T \rightarrow \infty} \mathbb{E} \|x_T - x^*\|^2 &\leq \frac{2\eta_H^2}{1 - \kappa_H^2} \left[\Sigma_K^2 + \frac{1 + \kappa_H^2}{1 - \kappa_H^2} (\varepsilon_K + \beta_K)^2 \right]. \end{aligned} \quad (25)$$

What the bound means. Theorem 3 is the grouped-trajectory counterpart of Theorem 2. After K RLOO tool updates and one independent RLOO reasoning update per round, the reasoning parameters approach x^* within a bounded long-run expected squared error. The bound separates the error from stopping tool adaptation after K steps (ε_K), systematic error from composing sampled updates (β_K), and sampling variation (Σ_K).

Recovering the deterministic K -step gradient.

Proposition 3 (Fixed- K deterministic-gradient recovery). *Under Assumption 5 (bounded unbiased group statistics on a compact path), fix $K < \infty$. For sequence n , take the large-batch limit at every inner step and at the independent task evaluation:*

$$N_{q,L,s}^{(n)} \rightarrow \infty \quad (s = 0, \dots, K-1), \quad N_{q,H}^{(n)} \rightarrow \infty, \quad (26)$$

where fixed $G \geq 2$ is allowed. Then

$$\sup_{x \in X} \mathbb{E} \|\hat{g}_K^{(n), \text{RLOO}}(x) - g_K(x)\|^2 \rightarrow 0. \quad (27)$$

For fixed K , increasing query counts at every inner step makes each sampled tool state and sensitivity approach its deterministic counterpart. Increasing the independent task query count then makes the grouped estimate approach g_K , the deterministic K -step gradient. Thus $\beta_{K,n}, \Sigma_{K,n} \rightarrow 0$, leaving only the finite-adaptation error ε_K from Section 5.2. This recovers the second level of the hierarchy but requires per-round query cost to grow at every adaptation and evaluation step. At finite batch sizes, Theorem 3 gives the corresponding error in terms of systematic and sampling effects. Its squared-error bound recovers Theorem 2’s finite-response radius within a factor in $[1, 2]$. For fixed $c < 2$ satisfying $\eta_H L_F^2 \leq c\mu_H$, this bound is $\Theta(\varepsilon_K^2/\mu_H^2)$. Appendix F.6 proves the comparison and its stated step-size dependence.

7 CONCLUSION

ECHO’s response-aware training treats reasoning and tool behavior as a single adaptive system. Its core contribution is a three-level analytical hierarchy linking practical GRPO to the Stackelberg objective. Exact response-aware updates converge to the equilibrium. K exact-gradient tool updates have bounded errors that vanish with K . Under stated conditions, grouped trajectory updates have bounded stochastic error and recover the deterministic K -step gradient as each update and evaluation uses a larger query batch.

Reproducibility statement. The empirical section will report the models, datasets, prompts, training settings, random seeds, and evaluation protocol needed to reproduce its results.

Ethics statement. Tool agents require permissioning, sandboxing, provenance checks, reliable evaluation, human review, and bias checks for training queries, rewards, and automated verifiers.

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A NOTATION

This appendix collects the notation used in the algorithm and analysis.

Symbol	Meaning
$x \in X \subset \mathbb{R}^{d_x}$	reasoning-policy leader parameter
$y \in Y \subset \mathbb{R}^{d_y}$	tool-policy follower parameter
$q \sim P_Q$	query and its fixed distribution
m_h, a_h, o_h	reasoning segment, tool action, and observation at step h
T_{ep}	episode horizon
$u_L(x, y), u_H(x, y)$	expected follower and leader utilities
$\xi(x)$	exact follower response
$y^{(s)}(x), \xi_K(x)$	follower iterate and K -step response
$Z_s(x) = D_x y^{(s)}(x)$	response sensitivity after s steps
D_0, D_Z	initial response and sensitivity radii
$F(x), F_K(x)$	exact- and finite-response leader objectives
$g_{\text{ex}}(x), g_K(x)$	their gradients
κ_L, κ_H	inner and outer contraction factors
ε_K	deterministic finite-response gradient error
$B_i, G_i, i \in \{L, H\}$	practical GRPO query batches and rollouts per group
N_q, G	analytical RLOO query groups and rollouts per group
$\hat{U}^{\text{GRPO}}$	standardized, clipped practical GRPO surrogate
$\hat{L}^{\text{RLOO}}$	raw-return full-trajectory analytical surrogate
$\mathcal{S}_c, \mathcal{T}_{\text{cd}}$	trajectory score and score-Hessian tensor
$\hat{y}_s^{\text{RLOO}}, \hat{Z}_s^{\text{RLOO}}$	sampled RLOO response and sensitivity
$\hat{g}_{K,t}^{\text{RLOO}}$	composed RLOO outer oracle
β_K	conditional RLOO composition-bias bound
Σ_K^2	conditional outer-noise second-moment bound
$\mathfrak{B}_K$	total finite-response-plus-composition bias

B EXTENDED RELATED WORK

Joint reasoning–tool trajectories. Language-model agents often alternate between an internal reasoning segment and an external action such as search, retrieval, or computation. ReAct makes this reasoning–acting sequence explicit in prompted trajectories (Yao et al., 2023). Toolformer and ToolLLM develop methods for learning when and how to invoke tools (Schick et al., 2023; Qin et al., 2024), and ToRL studies reinforcement learning for integrated reasoning and tool use (Li et al., 2025). ECHO begins from the same joint trajectory but focuses on its learning dynamics. A change in reasoning alters the histories and tool requests used for tool-policy adaptation. The adapted tool policy then changes the observations that shape later reasoning and final task return. The resulting object is a reasoning policy evaluated through the tool response induced by its own commitment.

Response maps and finite adaptation. The leader–follower terminology expresses the order of the ECHO update: the reasoning policy commits, and the tool policy adapts under that commitment. The response derivative is closely connected to gradient-based meta-learning and bilevel optimization, where an outer criterion is differentiated through an inner learning procedure or solution (Finn et al., 2017; Franceschi et al., 2018). Reverse differentiation computes the transpose action of an optimization path without materializing its full sensitivity matrix (Maclaurin et al., 2015). Analyses of truncated and iterative hypergradients quantify how a finite inner path approximates a response (Shaban et al., 2019; Grazi et al., 2020; Ji et al., 2021). Convergence analyses also cover gradient-based meta-learning and stochastic hypergradient approximation (Fallah et al., 2020; Grazi et al., 2021).

ECHO uses these tools at a specific reasoning–tool interface. The inner utility trains tool behavior on trajectories generated under the reasoning commitment, while the outer utility evaluates final task performance after the tool update. The mixed derivative therefore carries a trajectory-level effect from reasoning into tool adaptation. The paper follows this effect through K updates, compares it with the exact response, and bounds the separation between stationary points selected by response-aware and frozen-response gradients. ECHO combines this gradient approximation with the standard smooth projected-gradient contraction (Nesterov, 2004) to obtain the outer-iterate neighborhood.

Language-model response hierarchies. Stackelberg formulations for language models use the learned response for several distinct purposes. STA-RLHF places a language-model policy above a preference-model response, and Gauthier et al. differentiate policy learning through reward-model retraining (Makar-Limanov et al., 2024; Gauthier et al., 2026). SLHF conditions a follower’s refinement on a leader-generated candidate (Pásztor et al., 2026). Bi-NAC instead makes a feedback-generating critic the leader and optimizes it through the adaptation of a reasoning actor (Singh et al., 2026). Resource-aware agents use a controller commitment over quality and cost incentives to elicit an executor’s choices over context, prompting, and tool access (Wang, 2026). ECHO models two policy roles within one joint trajectory. The reasoning commitment changes the tool policy’s adaptation data, and the reasoning update differentiates through the resulting K -step response.

Practical grouped updates. GRPO forms query-conditioned rollout groups, standardizes returns within each group, and applies a clipped likelihood-ratio objective (Shao et al., 2024). Algorithm 1 uses this update in two phases. The follower phase adapts the tool policy using its role-specific return. A fresh task-evaluation group in the leader phase then supplies masked task-return GRPO directions for the reasoning policy and the adapted tool policy, and the latter is propagated backward through the tool updates. Token masks route these role-specific losses in a shared-model implementation. Separate policies provide the corresponding modular realization.

RLOO for first- and second-order response derivatives. The multiple-sample leave-one-out REINFORCE baseline centers each rollout by the mean return of the other rollouts generated for the same input (Kool et al., 2019). Its independence from the focal rollout gives the standard first-order score identity, and sequence-level RLOO has been studied for language-model optimization (Ahmadian et al., 2024). Our RLOO analysis uses this query-relative construction for both derivative orders needed by the response calculation. With a full-trajectory ratio, differentiating the RLOO surrogate yields the local policy-gradient block and the yy and yx score tensors that drive the follower Jacobian and its dependence on the reasoning commitment.

This calculation is connected to higher-order stochastic-computation-graph methods. DiCE constructs repeatedly differentiable Monte Carlo objectives (Foerster et al., 2018). Later work develops control variates and general stochastic automatic-differentiation recipes for higher-order estimates (Mao et al., 2019; van Krieken et al., 2021). Meta-gradient policy optimization likewise differentiates through policy adaptation (Tang et al., 2021). The ECHO derivation specializes these ideas to the complete joint trajectory and exposes the cross-role score product in the mixed Hessian. The resulting first- and second-order identities provide the local blocks required by the reverse response sweep.

RLOO insight for response-aware GRPO. The stochastic theorem uses fresh batches, raw sequence returns, a leave-one-out baseline, full-trajectory likelihood ratios, and role-isolated trainable coordinates. These conditions give one expected-utility target for the local first- and second-order calculations. Query-conditioned groups, within-query relative credit, role-routed derivatives, and propagation through the K -step tool response connect this analytical model to Algorithm 1. The RLOO results thereby provide first- and second-order insight for the practical response-aware GRPO procedure, which uses standardized and clipped updates.

C GRADIENT COMPUTATION FOR ECHO

This appendix records the exact and finite-response derivatives, the three GRPO surrogates used by Algorithm 1, and the reverse computation of its response term. Both utilities use argument order (x, y) . The reasoning policy controls x , and the tool policy controls y .

C.1 MODEL AND IMPLEMENTATION CONVENTIONS

Separate reasoning and tool policies instantiate x and y directly. A shared realization instead uses

$$w(x, y) = w_{\text{core}} + P_H x + P_L y, \quad (28)$$

where w_{core} and the routing maps are fixed, every trainable coordinate is represented by x or y , and the ranges of P_H and P_L may overlap. Role/token masks select the generated positions entering each loss. The routing maps determine which parameter block receives that loss. Shared physical weights

can therefore couple both role-conditioned policies. At outer round t , the implementation retains x_t , resets y and its follower optimizer state, evaluates working copies $w(x_t, \hat{y}_s)$, and discards the adapted y copy after the reasoning update. Thus the shared implementation has the same common-base response semantics as the separate one. A stateful follower optimizer is included in the follower state, reset at the same point, and differentiated through.

For the expected utilities, entropy and forward KL may serve different modeling purposes even though, for a fixed full-support reference,

$$-D_{\text{KL}}(p \parallel p_{\text{ref}}) = \text{Ent}(p) + \mathbb{E}_p[\log p_{\text{ref}}].$$

Curvature accounting therefore counts their shared entropy component once. The theory states strong concavity directly as a condition on u_{L} .

C.2 EXACT AND FINITE RESPONSE DERIVATIVES

At the unique interior response $y = \xi(x)$, follower stationarity gives

$$\nabla_y u_{\text{L}}(x, \xi(x)) = 0. \quad (29)$$

With the mixed-block convention $\nabla_{yx}^2 u_{\text{L}} := D_x(\nabla_y u_{\text{L}})$, differentiation of Eq. (29) yields

$$\nabla_{yy}^2 u_{\text{L}}(x, \xi(x)) D_x \xi(x) + \nabla_{yx}^2 u_{\text{L}}(x, \xi(x)) = 0.$$

Strong concavity makes the first matrix nonsingular, hence

$$D_x \xi(x) = -[\nabla_{yy}^2 u_{\text{L}}(x, \xi(x))]^{-1} \nabla_{yx}^2 u_{\text{L}}(x, \xi(x)).$$

The chain rule for $F(x) = u_{\text{H}}(x, \xi(x))$ gives

$$\nabla F(x) = \nabla_x u_{\text{H}}(x, \xi(x)) + D_x \xi(x)^\top \nabla_y u_{\text{H}}(x, \xi(x)).$$

For the deterministic finite response, write

$$y^{(s+1)}(x) = y^{(s)}(x) + \eta_{\text{L}} \nabla_y u_{\text{L}}(x, y^{(s)}(x)), \quad y^{(0)} = y_{\text{init}}.$$

If $Z_s(x) := D_x y^{(s)}(x)$, direct differentiation gives

$$Z_{s+1} = (I + \eta_{\text{L}} H_{yy,s}) Z_s + \eta_{\text{L}} H_{yx,s}, \quad Z_0 = 0,$$

where $H_{yy,s} := \nabla_{yy}^2 u_{\text{L}}(x, y^{(s)}(x))$ and $H_{yx,s} := \nabla_{yx}^2 u_{\text{L}}(x, y^{(s)}(x))$. Thus, for $F_K(x) := u_{\text{H}}(x, y^{(K)}(x))$,

$$g_K(x) := \nabla F_K(x) = \nabla_x u_{\text{H}}(x, y^{(K)}(x)) + Z_K(x)^\top \nabla_y u_{\text{H}}(x, y^{(K)}(x)).$$

Already at $K = 1$, differentiating $u_{\text{H}}(x, y_{\text{init}} + \eta_{\text{L}} \nabla_y u_{\text{L}}(x, y_{\text{init}}))$ produces the direct and response terms. We can differentiate the composition once or add the response term once to the direct partial.

C.3 GRPO COMPONENTS IN THE FOLLOWER AND LEADER PHASES

For an indexed query batch $\mathcal{Q} = (q_b)_{b=1}^B$ with $q_b \stackrel{\text{iid}}{\sim} P_{\mathcal{Q}}$, let $\mathcal{G}_b = (\tau_b^{(j)})_{j=1}^G$ be the rollout group generated while q_b is fixed. Let $\mathcal{R}_i$ be the realized return for $i \in \{\text{L}, \text{H}\}$ and let $p \in \{\text{L}, \text{H}\}$ denote the role whose token probabilities receive that return. The index set $\mathcal{I}_p(\tau_b^{(j)})$ contains the token positions assigned to role p by the loss mask. An empty mask gives zero gradient for this token loss.

GRPO uses the standardized within-query advantage

$$\hat{A}_{i,b}^{(j), \text{GRPO}} = \frac{\mathcal{R}_{i,b}^{(j)} - \bar{\mathcal{R}}_{i,b}}{\hat{\sigma}_{i,b} + \epsilon_{\text{adv}}}, \quad \bar{\mathcal{R}}_{i,b} := \frac{1}{G} \sum_{k=1}^G \mathcal{R}_{i,b}^{(k)}, \quad (30)$$

where $\hat{\sigma}_{i,b}$ is the standard deviation of the G returns for query q_b . For a token $z_{b,\ell}^{(j)}$ with history $c_{b,\ell}^{(j)}$, the likelihood ratio to the behavior snapshot is

$$w_{p,b,\ell}^{(j), \text{tok}} = \frac{\pi_{p,\vartheta_p}(z_{b,\ell}^{(j)} \mid c_{b,\ell}^{(j)})}{\pi_{p,\vartheta_p^{\text{old}}}(z_{b,\ell}^{(j)} \mid c_{b,\ell}^{(j)})}, \quad \vartheta_{\text{H}} = x, \quad \vartheta_{\text{L}} = y.$$

With $\text{clip}(a, l, u) := \min\{u, \max\{l, a\}\}$, the generic masked surrogate is

$$\begin{aligned} & \text{GRPO}(\mathcal{R}_i, \pi_p; \mathcal{Q}, \mathcal{G}) \\ &:= \frac{1}{BG} \sum_{b=1}^B \sum_{j=1}^G \frac{1}{\max\{1, |\mathcal{I}_p(\tau_b^{(j)})|\}} \sum_{\ell \in \mathcal{I}_p(\tau_b^{(j)})} \min\left\{w_{p,b,\ell}^{(j),\text{tok}} \hat{A}_{i,b}^{(j),\text{GRPO}}, \right. \\ & \quad \left. \text{clip}(w_{p,b,\ell}^{(j),\text{tok}}, 1 - \epsilon_p, 1 + \epsilon_p) \hat{A}_{i,b}^{(j),\text{GRPO}}\right\} - \hat{\mathcal{K}}_{i,p}. \end{aligned} \quad (31)$$

Here $\hat{\mathcal{K}}_{i,p}$ is the allocated KL contribution for utility i and role- p parameters. Each KL or entropy regularizer enters the sampled objective exactly once: either through this explicit term or through its designated return and derivative.

Using $\mathcal{G}_{L,t}^{(s)} = (\mathcal{G}_{L,t,b}^{(s)})_{b=1}^{B_L}$ and $\mathcal{G}_{H,t} = (\mathcal{G}_{H,t,b})_{b=1}^{B_H}$, the three ECHO instances are

$$\begin{aligned} \hat{U}_{L,t,s}^{\text{GRPO}} &= \text{GRPO}(\mathcal{R}_L, \pi_{L,y}; \mathcal{Q}_{L,t}^{(s)}, \mathcal{G}_{L,t}^{(s)}), \\ \hat{U}_{H,t}^{\text{reason}} &= \text{GRPO}(\mathcal{R}_H, \pi_{H,x}; \mathcal{Q}_{H,t}, \mathcal{G}_{H,t}), \\ \hat{U}_{H,t}^{\text{tool}} &= \text{GRPO}(\mathcal{R}_H, \pi_{L,y}; \mathcal{Q}_{H,t}, \mathcal{G}_{H,t}). \end{aligned} \quad (32)$$

The first surrogate adapts the tool policy using the tool-level return. The second gives the direct masked reasoning-token GRPO direction under the task-level return. The third gives the corresponding tool-token direction. Propagating it through the K updates produces the response term. With disjoint role coordinates, these directions are the respective role partial estimators.

The displayed protocol takes one update on each fresh follower-phase group. Its behavior snapshot generated that group, so the token ratio equals one and clipping is locally inactive at the displayed gradient. If a group is reused, each clipped optimizer epoch and its fixed behavior snapshot is a separate substep of the adaptation map. The main equations show stateless ascent. With momentum or Adam, the algorithm augments the follower state, resets y to y_{init} and the optimizer state to its initial value, and differentiates through the optimizer during the reverse sweep.

For separate trainable policies, $\mathcal{I}_H$ and $\mathcal{I}_L$ route the two losses to their respective policy blocks. For a shared trainable model, they remain role/token loss masks over one parameter vector. The masks select loss positions, while the shared weights can transmit either masked loss to both role-conditioned behaviors. In that realization, π_{p,ϑ_p} in Eq. (31) is the role-conditioned view $\pi_p(\cdot; w(x, y))$, followed by the routed x - or y -derivative specified by P_H or P_L .

C.4 PARTIAL-VERSUS-TOTAL DERIVATIVE AND REVERSE ADJOINT

The practical response calculation differentiates a GRPO update whose rollout group is policy generated. For the one-update protocol, the target and behavior policies coincide and the group contribution to the tool direction is

$$\hat{g}_{L,s,b}^{\text{grp}} := \frac{1}{G_L} \sum_{j=1}^{G_L} \frac{\hat{A}_{L,b}^{(j),\text{GRPO}}}{\max\{1, |\mathcal{I}_L(\tau_b^{(j)})|\}} \sum_{\ell \in \mathcal{I}_L(\tau_b^{(j)})} \nabla_y \log \pi_L(z_{b,\ell}^{(j)} | c_{b,\ell}^{(j)}; x, y), \quad (33)$$

$$\mathcal{S}_{c,s,b}^{\text{grp}} := \sum_{j=1}^{G_L} \nabla_c \log p_{x,y}(\tau_{L,t,b}^{(s,j)} | q_{L,t,b}^{(s)}), \quad c \in \{x, y\}. \quad (34)$$

Both expressions are evaluated at $(x_t, \hat{y}_s)$. The group score covers every stochastic reasoning and tool token in the complete query group. The derivative D_c^{fix} below holds the realized trajectories, returns, and standardized advantages fixed while differentiating the displayed score-vector expression. If $g_{L,s}^{\text{ex}}$ collects an exactly evaluated regularizer gradient, then $\hat{g}_{L,s}^{\text{GRPO}} = B_L^{-1} \sum_b \hat{g}_{L,s,b}^{\text{grp}} + g_{L,s}^{\text{ex}}$. An empirically evaluated regularizer is instead included in the corresponding group contribution and score correction, so its derivative is counted once.

For $c \in \{x, y\}$, the higher-order score-corrected GRPO block is

$$\hat{H}_{y,c,s}^{\text{GRPO}} := \frac{1}{B_L} \sum_{b=1}^{B_L} \left[D_c^{\text{fix}} \hat{g}_{L,s,b}^{\text{grp}} + \text{stopgrad}(\hat{g}_{L,s,b}^{\text{grp}})(\mathcal{S}_{c,s,b}^{\text{grp}})^\top \right] + D_c g_{L,s}^{\text{ex}}. \quad (35)$$

The first term differentiates the fixed-rollout GRPO computation. The score term carries the dependence of the query-group law on the current reasoning and tool coordinates. Thus, for separate policies, the fixed-rollout x -derivative can vanish while the trajectory-score term still conveys how a reasoning commitment changes the data used by the tool update. Under shared weights, both terms can contribute to the corresponding Jacobian.

Conditionally on the current state and before drawing the fresh query groups, under common-support and domination conditions that permit differentiation under the expectation, the parameter-independent query and reward laws give

$$\mathbb{E}[\hat{H}_{yc,s}^{\text{GRPO}} | x_t, \hat{y}_s] = D_c \mathbb{E}[\hat{g}_{L,s}^{\text{GRPO}}(x, y)]|_{(x_t, \hat{y}_s)}, \quad c \in \{x, y\}.$$

Hence Eq. (35) is a score-corrected Jacobian estimator of the mean on-policy GRPO direction.

Vector products evaluate these blocks implicitly. Holding a vector v fixed gives

$$\begin{aligned} (\hat{H}_{yc,s}^{\text{GRPO}})^\top v &= \frac{1}{B_L} \sum_{b=1}^{B_L} \left[\nabla_c^{\text{fix}} \langle \hat{g}_{L,s,b}^{\text{grp}}, v \rangle + \mathcal{S}_{c,s,b}^{\text{grp}} \text{stopgrad} \langle \hat{g}_{L,s,b}^{\text{grp}}, v \rangle \right] \\ &\quad + (D_c \hat{g}_{L,s}^{\text{ex}})^\top v. \end{aligned} \quad (36)$$

This is the score-corrected Jacobian-transpose–vector/vector–Jacobian construction used by the practical reverse sweep. Section 6 establishes the stochastic identities for the separately defined RLOO path used in the convergence analysis.

The score-corrected forward sensitivity is

$$\hat{Z}_{s+1} = (I + \eta_L \hat{H}_{yy,s}^{\text{GRPO}}) \hat{Z}_s + \eta_L \hat{H}_{yx,s}^{\text{GRPO}}, \quad \hat{Z}_0 = 0. \quad (37)$$

Put $A_s := I + \eta_L \hat{H}_{yy,s}^{\text{GRPO}}$ and $B_s := \eta_L \hat{H}_{yx,s}^{\text{GRPO}}$. Recursion (37) gives

$$\hat{Z}_K = \sum_{r=0}^{K-1} A_{K-1} A_{K-2} \cdots A_{r+1} B_r,$$

where an empty product is the identity. For $v_K = \hat{g}_t^{\text{fol}}$ and $v_s = A_s^\top v_{s+1}$, $s = K-1, \dots, 1$,

$$\hat{Z}_K^\top \hat{g}_t^{\text{fol}} = \sum_{r=0}^{K-1} B_r^\top v_{r+1} = \eta_L \sum_{r=0}^{K-1} (\hat{H}_{yx,r}^{\text{GRPO}})^\top v_{r+1},$$

which is Eq. (11). The equality is exact for the realized local blocks and is computed with Jacobian-transpose–vector or vector–Jacobian products.

C.5 FULL-TRAJECTORY LIKELIHOOD-RATIO DERIVATIVES FOR THE RLOO ANALYSIS

The analytical RLOO parameterization uses role-isolated formal coordinates:

$$D_y \log \pi_H(\cdot | h; x, y) = 0, \quad D_x \log \pi_L(\cdot | h; x, y) = 0. \quad (38)$$

Separate policies or disjoint trainable adapters realize this parameterization. Algorithm 1’s practical shared model instead allows the two role-conditioned policies to share weights.

For the analytical expected-utility target in Eq. (59), write $\ell_\vartheta(\tau | q) := \log p_\vartheta(\tau | q)$. The score and second likelihood-ratio derivative are

$$\mathcal{S}_c = \nabla_c \ell_\vartheta, \quad \mathcal{T}_{cd} = D_d \mathcal{S}_c + \mathcal{S}_c \mathcal{S}_d^\top.$$

The common-support and domination assumptions imply

$$\begin{aligned} \mathbb{E}[\mathcal{S}_c | q] &= 0, & \mathbb{E}[\mathcal{T}_{cd} | q] &= 0, \\ \nabla_c U_i(\vartheta) &= \mathbb{E}[\mathcal{R}_i \mathcal{S}_c] + \nabla_c \mathcal{J}_i(\vartheta), & D_d \nabla_c U_i(\vartheta) &= \mathbb{E}[\mathcal{R}_i \mathcal{T}_{cd}] + D_d \nabla_c \mathcal{J}_i(\vartheta). \end{aligned} \quad (39)$$

The two follower blocks used by response-aware unrolling are

$$\mathcal{T}_{yy} = \nabla_{yy}^2 \ell_\vartheta + \mathcal{S}_y \mathcal{S}_y^\top, \quad \mathcal{T}_{yx} = D_x \mathcal{S}_y + \mathcal{S}_y \mathcal{S}_x^\top.$$

The full-trajectory ratio in Eq. (60) produces both the first- and second-order blocks under ordinary autodifferentiation. In particular, the cross-role score product records how reasoning changes the trajectories on which the tool response is learned. Section 6 uses these identities to obtain first- and second-order insight for the practical response-aware GRPO procedure.

D RESPONSE AND SENSITIVITY BOUNDS

This appendix derives the exact response Jacobian and the deterministic finite-response error used in Section 5. Every utility is written in the common argument order (x, y) .

D.1 EXACT RESPONSE AND IMPLICIT SENSITIVITY

Lemma 1 (Exact follower response and sensitivity). *Under Assumption 1, the follower response is unique and continuously differentiable. Moreover,*

$$\|\xi(x) - \xi(x')\| \leq \frac{M_{yx}}{\mu_L} \|x - x'\|, \quad (40)$$

$$D\xi(x) = -[\nabla_{yy}^2 u_L(x, \xi(x))]^{-1} \nabla_{yx}^2 u_L(x, \xi(x)), \quad \|D\xi(x)\| \leq \frac{M_{yx}}{\mu_L}. \quad (41)$$

Proof. Compactness gives existence and strong concavity gives uniqueness. Because $\xi(x) \in \text{int}(Y)$, ordinary first-order stationarity gives

$$\nabla_y u_L(x, \xi(x)) = 0. \quad (42)$$

Strong concavity and smoothness imply

$$-L_L I \preceq \nabla_{yy}^2 u_L(x, y) \preceq -\mu_L I \quad \text{on } X \times Y. \quad (43)$$

The follower Hessian at the response is therefore invertible, with inverse norm at most $1/\mu_L$. The implicit function theorem applied on the open neighborhood from Assumption 1 supplies a local C^1 stationary response around every $x \in X$. Uniqueness makes these local maps agree on overlaps. Differentiating Eq. (42) yields

$$\nabla_{yy}^2 u_L(x, \xi(x)) D\xi(x) + \nabla_{yx}^2 u_L(x, \xi(x)) = 0,$$

which proves Eq. (41) and its norm bound.

For the global Lipschitz claim, put $y = \xi(x)$ and $y' = \xi(x')$. Strong concavity of $u_L(x, \cdot)$ and stationarity give

$$\begin{aligned} \mu_L \|y - y'\|^2 &\leq \langle \nabla_y u_L(x, y') - \nabla_y u_L(x, y), y - y' \rangle \\ &\leq M_{yx} \|x - x'\| \|y - y'\|. \end{aligned}$$

The second line follows by integrating the mixed derivative along the segment from x' to x , which remains in X by convexity. Cancelling $\|y - y'\|$ when nonzero proves Eq. (40). When $\|y - y'\| = 0$, the same bound holds trivially. $\square$

Proposition 4 (Response-aware gradient). *Under Assumption 1,*

$$g_{\text{ex}}(x) := \nabla F(x) = \nabla_x u_H(x, \xi(x)) + D\xi(x)^\top \nabla_y u_H(x, \xi(x)) \quad (44)$$

$$= \nabla_x u_H - (\nabla_{yx}^2 u_L)^\top (\nabla_{yy}^2 u_L)^{-\top} \nabla_y u_H, \quad (45)$$

where every derivative in the second line is evaluated at $(x, \xi(x))$.

Proof. Apply the chain rule to $F(x) = u_H(x, \xi(x))$, then substitute Eq. (41). The matrix dimensions are $D\xi \in \mathbb{R}^{d_y \times d_x}$, $\nabla_{yx}^2 u_L \in \mathbb{R}^{d_y \times d_x}$, and $\nabla_y u_H \in \mathbb{R}^{d_y}$. Both displayed expressions therefore belong to $\mathbb{R}^{d_x}$, the space of outer gradients. $\square$

The main convergence theorem assumes L_F directly. For completeness, the following optional lemma records one set of primitive conditions sufficient for that assumption. Strong concavity of F remains the separate condition in Assumption 2.

Lemma 2 (A sufficient reduced-smoothness bound). *In addition to Assumption 1, suppose the follower blocks are jointly Lipschitz on $X \times Y$:*

$$\begin{aligned} \|\nabla_{yy}^2 u_L(x, y) - \nabla_{yy}^2 u_L(x', y')\| &\leq \bar{\ell}_{yy} (\|x - x'\| + \|y - y'\|), \\ \|\nabla_{yx}^2 u_L(x, y) - \nabla_{yx}^2 u_L(x', y')\| &\leq \bar{\ell}_{yx} (\|x - x'\| + \|y - y'\|). \end{aligned} \quad (46)$$

Let $L_{Hxx}, L_{Hxy}, L_{Hyx}$ uniformly bound the corresponding Hessian blocks of u_H , set $L_\xi = M_{yx}/\mu_L$, and use $\bar{g}_H$ from Eq. (15). Then

$$L_{D\xi} := \frac{1 + L_\xi}{\mu_L} \left(\bar{\ell}_{yx} + \frac{M_{yx}}{\mu_L} \bar{\ell}_{yy} \right), \quad (47)$$

$$\|\nabla F(x) - \nabla F(x')\| \leq (L_{Hxx} + 2L_\xi L_{Hxy} + L_\xi^2 L_{Hyx} + \bar{g}_H L_{D\xi}) \|x - x'\|. \quad (48)$$

Proof. Write $H(x) = \nabla_{yy}^2 u_L(x, \xi(x))$ and $C(x) = \nabla_{yx}^2 u_L(x, \xi(x))$. The inverse identity and Eq. (41) give

$$\|D\xi(x) - D\xi(x')\| \leq \frac{1}{\mu_L} \|C(x) - C(x')\| + \frac{M_{yx}}{\mu_L^2} \|H(x) - H(x')\|.$$

Equation (40) and the two joint Lipschitz bounds prove Eq. (47). Next split the difference of

$$\nabla F(x) = \nabla_x u_H(x, \xi(x)) + D\xi(x)^\top \nabla_y u_H(x, \xi(x)).$$

The direct term has Lipschitz constant $L_{Hxx} + L_\xi L_{Hxy}$. Add and subtract $D\xi(x')^\top \nabla_y u_H(x, \xi(x))$ in the response term. Its two pieces have constants $\bar{g}_H L_{D\xi}$ and $L_\xi (L_{Hxy} + L_\xi L_{Hyx})$, respectively. Their sum is Eq. (48). $\square$

D.2 FINITE RESPONSE AND SENSITIVITY

Recall the common-base recursion

$$\begin{aligned} y^{(0)}(x) &= y_{\text{init}}, & y^{(s+1)}(x) &= y^{(s)}(x) + \eta_L \nabla_y u_L(x, y^{(s)}(x)), \\ \xi_K(x) &= y^{(K)}(x), & Z_s(x) &:= D_x y^{(s)}(x), \quad Z_0 = 0. \end{aligned}$$

Differentiating one update gives, in the displayed order,

$$Z_{s+1} = (I + \eta_L \nabla_{yy}^2 u_L(x, y^{(s)})) Z_s + \eta_L \nabla_{yx}^2 u_L(x, y^{(s)}). \quad (49)$$

Set $D_Z := \sup_x \|D\xi(x)\| \leq M_{yx}/\mu_L$ and define the boundary-safe finite convolution

$$\Gamma_K(\kappa) := \begin{cases} 0, & K = 0, \\ K\kappa^{K-1}, & K \geq 1, 0 < \kappa < 1, \\ 1, & K = 1, \kappa = 0, \\ 0, & K \geq 2, \kappa = 0. \end{cases}$$

For $\kappa_L = 1 - \eta_L \mu_L$, define

$$\begin{aligned} e_y(K) &:= D_0 \kappa_L^K, \\ C_Z &:= \eta_L \left(\ell_{yy} \frac{M_{yx}}{\mu_L} + \ell_{yx} \right) D_0, \\ e_Z(K) &:= D_Z \kappa_L^K + C_Z \Gamma_K(\kappa_L). \end{aligned}$$

Finally, recall

$$g_K(x) = \nabla_x u_H(x, \xi_K(x)) + Z_K(x)^\top \nabla_y u_H(x, \xi_K(x)).$$

Lemma 3 (Finite response, sensitivity, and hypergradient error). *Under Assumptions 1 and 3, for every integer $K \geq 0$, uniformly over $x \in X$,*

$$\|\xi_K(x) - \xi(x)\| \leq e_y(K), \quad (50)$$

$$\|Z_K(x) - D\xi(x)\| \leq e_Z(K), \quad (51)$$

$$\|g_K(x) - g_{\text{ex}}(x)\| \leq \varepsilon_K, \quad (52)$$

where

$$\varepsilon_K := L_{Hx,y} e_y(K) + \bar{g}_H e_Z(K) + \left(\frac{M_{yx}}{\mu_L} + e_Z(K) \right) L_{Hy,y} e_y(K). \quad (53)$$

For fixed $0 < \kappa_L < 1$, $\varepsilon_K = O(K\kappa_L^{K-1})$ and $\varepsilon_K \rightarrow 0$. If $\kappa_L = 0$, $e_y(K) = 0$ for $K \geq 1$ and $e_Z(K) = 0$ for $K \geq 2$.

Proof. Fix x and abbreviate $y^* = \xi(x)$ and $Z^* = D\xi(x)$. We first prove path containment and contraction together by induction. At $s = 0$, $y^{(0)} = y_{\text{init}} \in Y$. Whenever $y^{(s)} \in Y$, the segment from y^* to $y^{(s)}$ lies in Y , and stationarity plus the fundamental theorem of calculus gives

$$y^{(s+1)} - y^* = \left[I + \eta_L \int_0^1 \nabla_{yy}^2 u_L(x, y^* + r(y^{(s)} - y^*)) dr \right] (y^{(s)} - y^*). \quad (54)$$

By Eq. (43) and $\eta_L \leq 1/L_L$, the bracketed matrix has operator norm at most κ_L . Hence

$$\|y^{(s+1)} - y^*\| \leq \kappa_L \|y^{(s)} - y^*\| \leq D_0. \quad (55)$$

The tube condition places $y^{(s+1)}$ in Y , closing the induction. Iterating proves Eq. (50).

Put $H_s = \nabla_{yy}^2 u_L(x, y^{(s)})$ and $C_s = \nabla_{yx}^2 u_L(x, y^{(s)})$, with a star denoting evaluation at y^* . The implicit identity $H_* Z^* + C_* = 0$ is equivalent to

$$Z^* = (I + \eta_L H_*) Z^* + \eta_L C_*.$$

Subtracting it from Eq. (49), with $E_s = Z_s - Z^*$, gives

$$E_{s+1} = (I + \eta_L H_s) E_s + \eta_L (H_s - H_*) Z^* + \eta_L (C_s - C_*).$$

Using Assumption 3, Eq. (55), and $\|I + \eta_L H_s\| \leq \kappa_L$,

$$\|E_{s+1}\| \leq \kappa_L \|E_s\| + \eta_L \left(\ell_{yy} \frac{M_{yx}}{\mu_L} + \ell_{yx} \right) D_0 \kappa_L^s.$$

Since $Z_0 = 0$, $\|E_0\| \leq D_Z$. Unrolling in the written matrix-product order yields

$$\|E_K\| \leq D_Z \kappa_L^K + C_Z \sum_{s=0}^{K-1} \kappa_L^{K-1-s} \kappa_L^s = e_Z(K).$$

At $\kappa_L = 0$, the piecewise definition of Γ_K equals this finite sum and requires no division by κ_L .

For the leader gradient, abbreviate $p_x^K = \nabla_x u_H(x, \xi_K(x))$, $p_y^K = \nabla_y u_H(x, \xi_K(x))$, and use a star for exact-response evaluation. The exact three-term decomposition is

$$g_K - g_{\text{ex}} = (p_x^K - p_x^*) + (Z_K - Z^*)^\top p_y^* + Z_K^\top (p_y^K - p_y^*).$$

Furthermore, $\|Z_K\| \leq M_{yx}/\mu_L + e_Z(K)$. Applying Eqs. (17)–(18) and Eq. (15) proves Eq. (53).

For $0 < \kappa_L < 1$, the displayed envelopes have the stated order and vanish. If $\kappa_L = 0$, Eq. (54) reaches y^* after one step. The finite-convolution convention gives exact sensitivity after two. This proves every edge case in the statement. $\square$

E CONVERGENCE PROOFS FOR EXACT AND FINITE ADAPTATION

This appendix proves Theorems 1 and 2, followed by the response-term displacement proposition. The proof keeps the finite-adaptation error ε_K separate from any stochastic-oracle error.

Proof of Theorem 1. Strong concavity and L_F -smoothness imply, for all $x, x' \in X$,

$$\begin{aligned} \langle \nabla F(x) - \nabla F(x'), x - x' \rangle &\leq -\mu_H \|x - x'\|^2, \\ \|\nabla F(x) - \nabla F(x')\| &\leq L_F \|x - x'\|. \end{aligned}$$

The first-order optimality condition over the closed convex set X is equivalent, for every $\eta_H > 0$, to

$$x^* = \text{Proj}_X(x^* + \eta_H \nabla F(x^*)). \quad (56)$$

Here x^* is interior, so in fact $\nabla F(x^*) = 0$. Nonexpansiveness of Euclidean projection and $g_{\text{ex}} = \nabla F$ give

$$\begin{aligned} \|x_{t+1} - x^*\|^2 &\leq \|x_t - x^* + \eta_H (\nabla F(x_t) - \nabla F(x^*))\|^2 \\ &\leq (1 - 2\eta_H \mu_H + \eta_H^2 L_F^2) \|x_t - x^*\|^2 = \kappa_H^2 \|x_t - x^*\|^2. \end{aligned} \quad (57)$$

Taking square roots and iterating proves Eq. (12). If $\kappa_H = 0$, the same inequality states that the first update reaches x^* . No division by κ_H is needed.

Lemma 1 now gives

$$\|\xi(x_t) - \xi(x^*)\| \leq \frac{M_{yx}}{\mu_L} \|x_t - x^*\|,$$

which proves Eq. (13). Finally, smoothness and $\nabla F(x^*) = 0$ imply

$$0 \leq F(x^*) - F(x_t) \leq \frac{L_F}{2} \|x_t - x^*\|^2.$$

Substitution of Eq. (12) proves Eq. (14). $\square$

Proof of Theorem 2. Define $\Delta_K(x) := g_K(x) - g_{\text{ex}}(x)$. By Lemma 3, $\|\Delta_K(x)\| \leq \varepsilon_K$ uniformly. Combining Eq. (56), projection nonexpansiveness, the triangle inequality, and the exact contraction Eq. (57) yields

$$\begin{aligned} \|x_{t+1} - x^*\| &\leq \|x_t - x^* + \eta_H(g_{\text{ex}}(x_t) - g_{\text{ex}}(x^*))\| + \eta_H \|\Delta_K(x_t)\| \\ &\leq \kappa_H \|x_t - x^*\| + \eta_H \varepsilon_K. \end{aligned}$$

Unrolling this scalar recursion gives

$$\begin{aligned} \|x_t - x^*\| &\leq \kappa_H^t \|x_0 - x^*\| + \eta_H \varepsilon_K \sum_{j=0}^{t-1} \kappa_H^j \\ &= \kappa_H^t \|x_0 - x^*\| + \frac{\eta_H \varepsilon_K (1 - \kappa_H^t)}{1 - \kappa_H} = D_{t,K}, \end{aligned} \quad (58)$$

including $\kappa_H = 0$, for which the finite sum equals one whenever $t \geq 1$. This proves Eq. (20).

For the follower coordinate, add and subtract the exact response at x_t :

$$\begin{aligned} \|\xi_K(x_t) - \xi(x^*)\| &\leq \|\xi_K(x_t) - \xi(x_t)\| + \|\xi(x_t) - \xi(x^*)\| \\ &\leq e_y(K) + \frac{M_{yx}}{\mu_L} D_{t,K}, \end{aligned}$$

which is Eq. (21). Taking the upper limit in Eq. (58) proves Eq. (22).

For completeness, the same quantities give three utility bounds. Follower optimality, L_L -smoothness, and Eq. (50) imply

$$0 \leq u_L(x_t, \xi(x_t)) - u_L(x_t, \xi_K(x_t)) \leq \frac{L_L}{2} e_y(K)^2.$$

Outer smoothness gives

$$0 \leq F(x^*) - F(x_t) \leq \frac{L_F}{2} D_{t,K}^2.$$

Finally, the segment between $\xi_K(x_t)$ and $\xi(x_t)$ lies in Y , so the mean value theorem and Eq. (15) give

$$|u_H(x_t, \xi_K(x_t)) - F(x_t)| \leq \bar{g}_H e_y(K).$$

Thus

$$F(x^*) - u_H(x_t, \xi_K(x_t)) \leq \frac{L_F}{2} D_{t,K}^2 + \bar{g}_H e_y(K).$$

In particular, the limiting squared-distance contribution is at most $\eta_H^2 \varepsilon_K^2 / (1 - \kappa_H)^2$, making explicit the $O(\varepsilon_K^2)$ deterministic finite- K neighborhood. $\square$

Proof of Proposition 1. Apply strong concavity of F to $(x^\dagger, x^*)$ and use $\nabla F(x^*) = 0$:

$$\begin{aligned} \mu_H \|x^\dagger - x^*\|^2 &\leq -\langle \nabla F(x^\dagger), x^\dagger - x^* \rangle \\ &\leq \|\nabla F(x^\dagger)\| \|x^\dagger - x^*\|. \end{aligned}$$

If the two points differ, cancellation gives $\mu_H \|x^\dagger - x^*\| \leq \|\nabla F(x^\dagger)\|$. If they agree, the desired bound is immediate. By Proposition 4 and the defining condition $g^{\text{dir}}(x^\dagger) = 0$,

$$\nabla F(x^\dagger) = D\xi(x^\dagger)^\top \nabla_y u_H(x^\dagger, \xi(x^\dagger)).$$

This proves the first inequality in Eq. (16). The second follows from Lemma 1 and the definition of $\bar{g}_H$. The scalar quadratic example has a constant response term and reduced curvature exactly μ_H , so every inequality is an equality there. $\square$

F PROOFS FOR THE FULL-TRAJECTORY RLOO REFERENCE MODEL

This appendix proves the local derivative, finite-group, path-tracking, fixed- K consistency, and conditional transfer results from Section 6. The proof layer uses raw-return RLOO with full-trajectory likelihood ratios, fresh current-state batches, and role-isolated trainable coordinates. In particular, it verifies both the first-order gradient statistics and the second-order yy and yx statistics used by response-aware unrolling.

F.1 RLOO REFERENCE MODEL AND LOCAL REGULARITY

For $\vartheta = (x, y)$ and $i \in \{L, H\}$, use the utility-weighted returns $\mathcal{R}_L := \lambda_{\text{valid}} r_{\text{valid}}$ and $\mathcal{R}_H := r_{\text{task}}$ in this appendix. Set $\mathcal{J}_L(\vartheta) := \lambda_{\text{ent}} \mathcal{H}_{\text{tool}}(x, y) - \lambda_L \mathcal{K}_L(x, y)$ and $\mathcal{J}_H(\vartheta) := -\lambda_H \mathcal{K}_H(x, y)$, and define the expected-utility target

$$U_i(\vartheta) := \mathbb{E}_{q \sim P_Q, \tau \sim p_\vartheta(\cdot|q)}[\mathcal{R}_i(q, \tau)] + \mathcal{J}_i(\vartheta), \quad u_i(x, y) \equiv U_i(x, y). \quad (59)$$

Here $\mathcal{R}_i$ is parameter-independent, while $\mathcal{J}_i$ is the complete C^2 entropy-KL regularizer from Eqs. (2)–(3). Its exact first and second derivatives include all dependence through both the regularizer integrand and its parameter-dependent context law. The surrogate adds these complete derivatives directly, so its sampled and deterministic derivatives target the same utility.

At a detached behavior point $\bar{\vartheta}$, draw N_q independent queries and $G \geq 2$ conditionally independent trajectories per query. With $\bar{\mathcal{R}}_{i,b} := G^{-1} \sum_{k=1}^G \mathcal{R}_{i,b}^{(k)}$, set

$$\hat{A}_{i,b}^{(j),\text{LOO}} := \mathcal{R}_{i,b}^{(j)} - \frac{1}{G-1} \sum_{k \neq j} \mathcal{R}_{i,b}^{(k)} = \frac{G}{G-1} (\mathcal{R}_{i,b}^{(j)} - \bar{\mathcal{R}}_{i,b}),$$

$$w_{\vartheta, \bar{\vartheta}}(\tau | q) := \frac{p_\vartheta(\tau | q)}{p_{\bar{\vartheta}}(\tau | q)}, \quad (60)$$

$$\hat{\mathcal{L}}_i(\vartheta; \bar{\vartheta}, \mathcal{D}) := \frac{1}{N_q G} \sum_{b=1}^{N_q} \sum_{j=1}^G \text{stopgrad}(\hat{A}_{i,b}^{(j),\text{LOO}}) w_{\vartheta, \bar{\vartheta}}(\tau_b^{(j)} | q_b) + \mathcal{J}_i(\vartheta).$$

The reward-weighted part of the surrogate is zero at the behavior point within each group. Nevertheless, its local derivatives estimate those of U_i .

For derivative coordinates $c, d \in \{x, y\}$, write

$$\mathcal{S}_c := \nabla_c \log p_{x,y}(\tau | q),$$

$$\mathcal{T}_{cd} := D_d \mathcal{S}_c + \mathcal{S}_c \mathcal{S}_d^\top.$$

Assumption 4 (Local likelihood-ratio regularity). *The trajectory law is positive on parameter-independent common support and is twice differentiable with domination sufficient to exchange two derivatives and expectation. Query groups are independent, trajectories within a group are conditionally iid, Eq. (38) holds, and the same exact regularizer derivatives appear in the surrogate.*

F.2 LOCAL LOO IDENTITIES

Fix a behavior point $\bar{\vartheta}$ and condition on a query q . Suppress the utility and query indices. For a focal trajectory j , write

$$B_{-j} := \frac{1}{G-1} \sum_{k \neq j} \mathcal{R}^{(k)}, \quad \hat{A}^{(j),\text{LOO}} = \mathcal{R}^{(j)} - B_{-j}.$$

Conditional independence makes B_{-j} independent of every score tensor of trajectory j . Differentiating the likelihood ratio at $\vartheta = \bar{\vartheta}$ gives

$$\nabla_c w_{\vartheta, \bar{\vartheta}}|_{\bar{\vartheta}} = \mathcal{S}_c, \quad D_d \nabla_c w_{\vartheta, \bar{\vartheta}}|_{\bar{\vartheta}} = \mathcal{T}_{cd}.$$

The normalization identities in Eq. (39) therefore imply

$$\mathbb{E}[\hat{A}^{(j),\text{LOO}} \mathcal{S}_{c,j} | q] = \mathbb{E}[\mathcal{R} \mathcal{S}_c | q] - \mathbb{E}[B_{-j} | q] \mathbb{E}[\mathcal{S}_{c,j} | q] = \mathbb{E}[\mathcal{R} \mathcal{S}_c | q], \quad (61)$$

$$\mathbb{E}[\hat{A}^{(j),\text{LOO}} \mathcal{T}_{cd,j} | q] = \mathbb{E}[\mathcal{R} \mathcal{T}_{cd} | q]. \quad (62)$$

Averaging over j , then over independent query groups, and adding the exact derivatives of $\mathcal{J}_i$ proves Proposition 2. Both identities hold directly for every finite $G \geq 2$.

F.3 LOCAL FINITE-GROUP VARIANCE

For $\Psi = \mathcal{S}_c$ or $\Psi = \mathcal{T}_{cd}$, let $\mathcal{J}_{i,\Psi}$ denote, respectively, $\nabla_c \mathcal{J}_i$ or $D_d \nabla_c \mathcal{J}_i$, and define

$$\hat{D}_\Psi := \frac{1}{N_q G} \sum_{b=1}^{N_q} \sum_{j=1}^G \hat{A}_{i,b}^{(j),\text{LOO}} \Psi_{b,j} + \mathcal{J}_{i,\Psi}, \quad D_\Psi := \mathbb{E}[\mathcal{R}_i \Psi] + \mathcal{J}_{i,\Psi}.$$

Proposition 5 (RLOO finite-group mean-square bound). *Under Assumption 4, if $|\mathcal{R}_i| \leq R_i^{\max}$ and $\|\Psi\|_F \leq M_\Psi$ uniformly for $\Psi \in \{\mathcal{S}_c, \mathcal{T}_{cd}\}$, then the corresponding complete derivative $\hat{D}_\Psi$ satisfies*

$$\mathbb{E}\|\hat{D}_\Psi - D_\Psi\|_F^2 \leq \frac{V_{\Psi,\text{between}}}{N_q} + \frac{32(R_i^{\max})^2 M_\Psi^2}{N_q G}, \quad V_{\Psi,\text{between}} := \mathbb{E}_q \|m_\Psi(q) - \mathbb{E}_q m_\Psi(q)\|_F^2, \quad (63)$$

where $m_\Psi(q) = \mathbb{E}[\mathcal{R}_i \Psi \mid q]$.

To prove Eq. (63), condition on q and let

$$Z_\Psi := \frac{1}{G} \sum_{j=1}^G \hat{A}^{(j),\text{LOO}} \Psi_j.$$

Pairing the ordered terms gives the order-two U-statistic representation

$$Z_\Psi = \frac{1}{G(G-1)} \sum_{1 \leq j < k \leq G} (\mathcal{R}_j - \mathcal{R}_k)(\Psi_j - \Psi_k).$$

If one trajectory is replaced by an independent copy, only $G-1$ summands change. Each old or new summand has Frobenius norm at most $4R_i^{\max} M_\Psi$, so

$$\|Z_\Psi - Z_\Psi^{(\ell)}\|_F \leq \frac{8R_i^{\max} M_\Psi}{G}.$$

The Efron–Stein inequality in the Euclidean space obtained by vectorizing the statistic yields

$$\mathbb{E}[\|Z_\Psi - \mathbb{E}[Z_\Psi \mid q]\|_F^2 \mid q] \leq \frac{1}{2} \sum_{\ell=1}^G \mathbb{E}[\|Z_\Psi - Z_\Psi^{(\ell)}\|_F^2 \mid q] \leq \frac{32(R_i^{\max})^2 M_\Psi^2}{G}.$$

By Eqs. (61)–(62), $\mathbb{E}[Z_\Psi \mid q] = m_\Psi(q)$. The law of total variance and independence of the N_q query groups now give

$$\mathbb{E}\|\hat{D}_\Psi - D_\Psi\|_F^2 \leq \frac{V_{\Psi,\text{between}}}{N_q} + \frac{32(R_i^{\max})^2 M_\Psi^2}{N_q G},$$

which is the stated bound. Exact regularizer derivatives are nonrandom.

F.4 FOLLOWER-PATH TRACKING

At follower step s , instantiate a fresh $\hat{\mathcal{L}}_{L,s}^{\text{RLOO}}$ on independent formal inputs at $(x_t, \hat{y}_s^{\text{RLOO}})$ and define, for $c \in \{x, y\}$,

$$\begin{aligned} \hat{g}_{L,s}^{\text{RLOO}} &:= \nabla_y \hat{\mathcal{L}}_{L,s}^{\text{RLOO}}(x, y) \Big|_{(x_t, \hat{y}_s^{\text{RLOO}})}, & \hat{H}_{yc,s}^{\text{RLOO}} &:= D_c \nabla_y \hat{\mathcal{L}}_{L,s}^{\text{RLOO}}(x, y) \Big|_{(x_t, \hat{y}_s^{\text{RLOO}})}, \\ \hat{y}_{s+1}^{\text{RLOO}} &= \hat{y}_s^{\text{RLOO}} + \eta_L \hat{g}_{L,s}^{\text{RLOO}}, & \hat{Z}_{s+1}^{\text{RLOO}} &= (I + \eta_L \hat{H}_{yy,s}^{\text{RLOO}}) \hat{Z}_s^{\text{RLOO}} + \eta_L \hat{H}_{yx,s}^{\text{RLOO}}, \end{aligned} \quad (64)$$

with $\hat{y}_0^{\text{RLOO}} = y_{\text{init}}$ and $\hat{Z}_0^{\text{RLOO}} = 0$. On an independent task-evaluation batch, set

$$\hat{g}_t^{\text{dir,RLOO}} := \nabla_x \hat{\mathcal{L}}_{H,t}^{\text{RLOO}}(x, y) \Big|_{(x_t, \hat{y}_K^{\text{RLOO}})}, \quad \hat{g}_t^{\text{fol,RLOO}} := \nabla_y \hat{\mathcal{L}}_{H,t}^{\text{RLOO}}(x, y) \Big|_{(x_t, \hat{y}_K^{\text{RLOO}}}.$$

The composed RLOO oracle is

$$\hat{g}_{K,t}^{\text{RLOO}} := \hat{g}_t^{\text{dir,RLOO}} + (\hat{Z}_K^{\text{RLOO}})^\top \hat{g}_t^{\text{fol,RLOO}}. \quad (65)$$

Let $T_x(y) := y + \eta_L \nabla_y u_L(x, y)$ and $y^{(s+1)} = T_x(y^{(s)})$. With the fresh-batch filtration, define

$$\mathcal{F}_{t,s}^{\text{low}} := \sigma(\mathcal{F}_t, \mathcal{D}_{L,t,0}, \dots, \mathcal{D}_{L,t,s-1}).$$

Here $\mathcal{D}_{L,t,r}$ is the query-group data used to instantiate the fresh follower-phase RLOO surrogate at step r . Thus x_t , $\hat{y}_s^{\text{RLOO}}$, and $\hat{Z}_s^{\text{RLOO}}$ are measurable with respect to $\mathcal{F}_{t,s}^{\text{low}}$, while the step- s batch is conditionally fresh. Define

$$\zeta_{t,s+1}^L := \hat{g}_{L,s}^{\text{RLOO}} - \nabla_y u_L(x_t, \hat{y}_s^{\text{RLOO}}), \quad \mathbb{E}[\zeta_{t,s+1}^L \mid \mathcal{F}_{t,s}^{\text{low}}] = 0. \quad (66)$$

Proposition 6 (RLOO follower-path tracking). *If T_x is $\kappa_L < 1$ Lipschitz on a tube containing both paths and*

$$\mathbb{E}[\|\zeta_{t,s+1}^L\|^2 \mid \mathcal{F}_{t,s}^{\text{low}}] \leq \nu_s^2, \quad (67)$$

then, conditional on the pre-outer history,

$$\mathbb{E}\|\hat{y}_K^{\text{RLOO}} - y^{(K)}(x_t)\|^2 \leq \eta_L^2 \sum_{s=0}^{K-1} \kappa_L^{2(K-1-s)} \nu_s^2, \quad (68)$$

$$(\mathbb{E}\|\hat{y}_K^{\text{RLOO}} - \xi(x_t)\|^2)^{1/2} \leq e_y(K) + \eta_L \left(\sum_{s=0}^{K-1} \kappa_L^{2(K-1-s)} \nu_s^2 \right)^{1/2}. \quad (69)$$

Let $e_s := \hat{y}_s^{\text{RLOO}} - y^{(s)}(x_t)$. Equations (64) and (66) give

$$e_{s+1} = T_{x_t}(\hat{y}_s^{\text{RLOO}}) - T_{x_t}(y^{(s)}(x_t)) + \eta_L \zeta_{t,s+1}^L.$$

The first term is $\mathcal{F}_{t,s}^{\text{low}}$ -measurable, while the second has zero conditional mean. Thus the cross term vanishes after conditioning, and contractivity gives

$$\mathbb{E}[\|e_{s+1}\|^2 \mid \mathcal{F}_{t,s}^{\text{low}}] \leq \kappa_L^2 \|e_s\|^2 + \eta_L^2 \nu_s^2.$$

Since $e_0 = 0$, conditional iteration proves Eq. (68). Conditional Minkowski and $\|y^{(K)}(x_t) - \xi(x_t)\| \leq e_y(K)$ give Eq. (69).

Conditionally on the complete follower phase, freshness of the leader batch gives

$$\mathbb{E}[\hat{g}_{K,t}^{\text{RLOO}} \mid \mathcal{F}_{t,K}^{\text{low}}] = \nabla_x U_H(x_t, \hat{y}_K^{\text{RLOO}}) + (\hat{Z}_K^{\text{RLOO}})^\top \nabla_y U_H(x_t, \hat{y}_K^{\text{RLOO}}).$$

If almost-sure containment is to be obtained from a residual bound rather than assumed directly, one sufficient condition is $\|\zeta_{t,s+1}^L\| \leq \nu_{\max}$ almost surely and a regularity set containing the tube of radius

$$D_0 + \frac{\eta_L \nu_{\max}}{1 - \kappa_L} \quad (70)$$

around $\xi(X)$, where $D_0 := \sup_{x \in X} \|y_{\text{init}} - \xi(x)\|$. This follows by iterating the pathwise contraction inequality and preserves the unprojected follower recursion.

F.5 FIXED- K LARGE-BATCH CONSISTENCY

We spell out the assumptions used by Proposition 3.

Assumption 5 (Fixed- K large-batch regularity). *The initialization y_{init} and $Z_0 = 0$ are deterministic. The sets X and Y_{tube} are compact, and all deterministic and sampled follower states lie in Y_{tube} almost surely. At each state (x, y) , every follower-phase query-group statistic for g_L , H_{yy} , and H_{yx} is conditionally unbiased for its corresponding expected-utility derivative, is uniformly bounded, and has the declared current-state law determined by (x, y) . The leader-phase direct and follower partial statistics have the analogous properties and are conditionally fresh after the follower phase. Write $g_L := \nabla_y U_L$, $H_{yy} := D_y \nabla_y U_L$, and $H_{yx} := D_x \nabla_y U_L$. The utility-derivative maps*

$$g_L, \quad H_{yy}, \quad H_{yx}, \quad \nabla_x U_H, \quad \nabla_y U_H \quad (71)$$

are uniformly continuous on $X \times Y_{\text{tube}}$. These hypotheses include common support, exact role isolation, and every exact regularizer derivative.

When Eq. (70) supplies containment, Y_{tube} may be chosen as a compact regularity set containing that radius tube, which in turn contains $\mathcal{T}_{D_0}(\xi(X))$.

For the every-inner-step batch sequence in Eq. (26), the proof first establishes the stronger state-and-sensitivity conclusion, for every $s \leq K$,

$$\sup_{x \in X} \mathbb{E}[\|\hat{y}_s^{(n), \text{RLOO}} - y^{(s)}\|^2 + \|\hat{Z}_s^{(n), \text{RLOO}} - Z_s\|_{\text{F}}^2] \rightarrow 0. \quad (72)$$

We use the following elementary consequence repeatedly. If $\hat{M}_n(x, y)$ is an average of N_n conditionally independent, uniformly bounded, unbiased group statistics with mean $M(x, y)$, then

$$\sup_{(x, y) \in X \times Y_{\text{tube}}} \mathbb{E}\|\hat{M}_n(x, y) - M(x, y)\|_{\text{F}}^2 \leq \frac{C_M}{N_n} \rightarrow 0. \quad (73)$$

Uniformity follows from the common per-group second-moment bound, which also controls the supremum of the pointwise expectations.

We first prove state consistency. Put $e_s^{(n)}(x) := \hat{y}_s^{(n), \text{RLOO}}(x) - y^{(s)}(x)$. Given $e_s^{(n)} \rightarrow 0$ uniformly in mean square, write

$$\begin{aligned} e_{s+1}^{(n)} &= e_s^{(n)} + \eta_{\text{L}} \{ \hat{g}_{\text{L}, s}^{(n), \text{RLOO}}(x, \hat{y}_s^{(n), \text{RLOO}}) - g_{\text{L}}(x, \hat{y}_s^{(n), \text{RLOO}}) \\ &\quad + g_{\text{L}}(x, \hat{y}_s^{(n), \text{RLOO}}) - g_{\text{L}}(x, y^{(s)}) \}. \end{aligned}$$

The first braced difference converges uniformly in mean square by Eq. (73). For the second, uniform continuity and boundedness give convergence uniformly in mean square: for any $\delta > 0$, split on $\|e_s^{(n)}\| \leq \delta$ and use Markov's inequality on the complement, then send $n \rightarrow \infty$ and $\delta \downarrow 0$. Since $e_0^{(n)} = 0$, finite induction proves state consistency for every $s \leq K$.

For sensitivity, let $A_s(x, y) := I + \eta_{\text{L}} H_{yy}(x, y)$ and $B_s(x, y) := \eta_{\text{L}} H_{yx}(x, y)$, with hatted versions defined from the sampled blocks. Subtracting the deterministic and sampled recursions yields

$$\begin{aligned} \hat{Z}_{s+1}^{(n), \text{RLOO}} - Z_{s+1} &= \hat{A}_s^{(n), \text{RLOO}}(\hat{Z}_s^{(n), \text{RLOO}} - Z_s) + (\hat{A}_s^{(n), \text{RLOO}} - A_s(x, y^{(s)}))Z_s \\ &\quad + \hat{B}_s^{(n), \text{RLOO}} - B_s(x, y^{(s)}). \end{aligned}$$

Uniform boundedness controls the first multiplier and makes the deterministic Z_s uniformly bounded for fixed s . It also makes every sampled $\hat{Z}_s^{(n), \text{RLOO}}$ uniformly bounded almost surely for fixed s , by direct finite induction from $\hat{Z}_0^{(n), \text{RLOO}} = 0$. Equation (73), state consistency, and uniform continuity make the last two differences converge uniformly in mean square. Starting from $\hat{Z}_0^{(n), \text{RLOO}} = Z_0 = 0$, a second finite induction proves Eq. (72).

Finally decompose the task-evaluation oracle into its sampled partial-estimation error, the continuous leader-utility derivatives evaluated at $\hat{y}_K^{(n), \text{RLOO}}$ rather than $y^{(K)}$, and the product error from $\hat{Z}_K^{(n), \text{RLOO}} - Z_K$. The task-evaluation query count tending to infinity controls the first part by Eq. (73). State and sensitivity consistency plus uniform continuity and boundedness control the remaining parts. This proves Eq. (27). All inductions have exactly K steps. The conclusion therefore covers fixed K and a large-batch sequence in which the query count diverges at every inner step and during leader-phase task evaluation. The per-round sample cost grows along this sequence.

For completeness, Jensen's inequality and the variance decomposition imply from Eq. (27) that

$$\beta_{K,n}^2 := \sup_x \mathbb{E}\|\hat{g}_K^{(n), \text{RLOO}}(x) - g_K(x)\|^2 \leq \sup_x \mathbb{E}\|\hat{g}_K^{(n), \text{RLOO}}(x) - g_K(x)\|^2 \rightarrow 0, \quad (74)$$

$$\Sigma_{K,n}^2 := \sup_x \mathbb{E}\|\hat{g}_K^{(n), \text{RLOO}}(x) - \mathbb{E}\hat{g}_K^{(n), \text{RLOO}}(x)\|^2 \leq \sup_x \mathbb{E}\|\hat{g}_K^{(n), \text{RLOO}}(x) - g_K(x)\|^2 \rightarrow 0. \quad (75)$$

Fresh current-state sampling makes these uniform bounds conditional on $\mathcal{F}_t$ when evaluated at the random state x_t . They are therefore the constants used after Theorem 3.

F.6 CONDITIONAL STOCHASTIC-ORACLE TRANSFER

The conditional conditions in Theorem 3 are as follows. Let $x_0 \in X$ be deterministic and $\mathcal{F}_t$ be the pre-oracle history, so x_t is $\mathcal{F}_t$ -measurable and the current RLOO oracle is conditionally fresh. For every t , suppose almost surely that

$$\|\mathbb{E}[\hat{g}_{K,t}^{\text{RLOO}} | \mathcal{F}_t] - g_K(x_t)\| \leq \beta_K, \quad \mathbb{E}[\|\hat{g}_{K,t}^{\text{RLOO}} - \mathbb{E}[\hat{g}_{K,t}^{\text{RLOO}} | \mathcal{F}_t]\|^2 | \mathcal{F}_t] \leq \Sigma_K^2. \quad (76)$$

Proof of Theorem 3. Put

$$\mathfrak{B}_K := \varepsilon_K + \beta_K, \quad \tilde{\kappa}_H^2 := \frac{1 + \kappa_H^2}{2}, \quad \mathcal{C}_K := \Sigma_K^2 + \frac{1 + \kappa_H^2}{1 - \kappa_H^2} \mathfrak{B}_K^2.$$

The first-order optimality condition for maximizing the differentiable concave function F over the closed convex set X gives

$$x^* = \text{Proj}_X(x^* + \eta_H g_{\text{ex}}(x^*)). \quad (77)$$

Define the pre-projection exact displacement

$$d_t^{\text{pre}} := (x_t - x^*) + \eta_H (g_{\text{ex}}(x_t) - g_{\text{ex}}(x^*)). \quad (78)$$

Strong concavity and Lipschitz continuity give

$$\begin{aligned} \|d_t^{\text{pre}}\|^2 &\leq \|x_t - x^*\|^2 - 2\eta_H \mu_H \|x_t - x^*\|^2 + \eta_H^2 L_F^2 \|x_t - x^*\|^2 \\ &= \kappa_H^2 \|x_t - x^*\|^2. \end{aligned}$$

This pre-projection inequality, rather than only contraction of the projected map, is needed to eliminate the centered-noise cross term.

Let

$$\begin{aligned} b_t &:= \mathbb{E}[\hat{g}_{K,t}^{\text{RLOO}} | \mathcal{F}_t] - g_K(x_t), & \delta_t &:= g_K(x_t) - g_{\text{ex}}(x_t), \\ \zeta_{t+1}^H &:= \hat{g}_{K,t}^{\text{RLOO}} - \mathbb{E}[\hat{g}_{K,t}^{\text{RLOO}} | \mathcal{F}_t]. \end{aligned}$$

Then $\|b_t + \delta_t\| \leq \mathfrak{B}_K$, and the centered term has conditional second moment at most Σ_K^2 . Projection nonexpansiveness, Eqs. (77) and (78), and conditional centering yield

$$\mathbb{E}[\|x_{t+1} - x^*\|^2 | \mathcal{F}_t] \leq (\kappa_H \|x_t - x^*\| + \eta_H \mathfrak{B}_K)^2 + \eta_H^2 \Sigma_K^2.$$

For $0 < \kappa_H < 1$, apply Young's inequality to the cross term with $u = \kappa_H \|x_t - x^*\|$, $v = \eta_H \mathfrak{B}_K$, and $\rho = (1 - \kappa_H^2)/(2\kappa_H^2)$. Then $\kappa_H^2(1 + \rho) = (1 + \kappa_H^2)/2$ and $1 + \rho^{-1} = (1 + \kappa_H^2)/(1 - \kappa_H^2)$, which gives

$$\mathbb{E}[\|x_{t+1} - x^*\|^2 | \mathcal{F}_t] \leq \tilde{\kappa}_H^2 \|x_t - x^*\|^2 + \eta_H^2 \mathcal{C}_K.$$

For $\kappa_H = 0$, the cross term is zero and the same inequality follows directly. Taking total expectations and iterating the affine recursion gives

$$\mathbb{E}\|x_T - x^*\|^2 \leq \tilde{\kappa}_H^{2T} \|x_0 - x^*\|^2 + \eta_H^2 \frac{1 - \tilde{\kappa}_H^{2T}}{1 - \tilde{\kappa}_H^2} \mathcal{C}_K.$$

Taking the limit superior and using $1 - \tilde{\kappa}_H^2 = (1 - \kappa_H^2)/2$ proves Eq. (25). $\square$

Setting $\Sigma_K = 0$ and $\mathfrak{B}_K = \varepsilon_K$ in Eq. (25) gives

$$\frac{2\eta_H^2(1 + \kappa_H^2)}{(1 - \kappa_H^2)^2} \varepsilon_K^2 = \frac{2(1 + \kappa_H^2)}{(1 + \kappa_H)^2} \left(\frac{\eta_H \varepsilon_K}{1 - \kappa_H} \right)^2.$$

The prefactor $2(1 + \kappa_H^2)/(1 + \kappa_H)^2$ lies in $[1, 2]$. Moreover,

$$\frac{\eta_H}{1 - \kappa_H} = \frac{1 + \kappa_H}{2\mu_H - \eta_H L_F^2}.$$

Under $\eta_H L_F^2 \leq c\mu_H$ for fixed $c < 2$, the right-hand side lies between $1/(2\mu_H)$ and $2/((2 - c)\mu_H)$. Hence the *certified asymptotic term* in the large-batch limit is $\Theta(\varepsilon_K^2/\mu_H^2)$. Together with Eqs. (74)–(75), this shows that $\beta_{K,n} \rightarrow 0$, $\Sigma_{K,n} \rightarrow 0$, and $\mathfrak{B}_{K,n} \rightarrow \varepsilon_K$, recovering the finite-response scale of the deterministic finite-response theorem up to constants.

G SCALAR CHECKS AND A POLICY-COORDINATE WITNESS

We give two complementary deterministic checks. A quadratic family makes the finite- K errors and the response-gap bound exact. A Bernoulli probability-coordinate example then verifies that all domain, interiority, curvature, tube-containment, and step-size assumptions can hold in a nonlinear policy model. These compact examples provide algebraic validation of the deterministic theory.

G.1 QUADRATIC FAMILY: EXACT ERROR AND TIGHT DISPLACEMENT

For scalar x and y , consider

$$u_L(x, y) = -\frac{1}{2}(y - ax)^2, \quad u_H(x, y) = -\frac{1}{2}(x - b)^2 + cy.$$

Follower stationarity gives $\xi(x) = ax$ and $D\xi(x) = a$, so

$$F(x) = -\frac{1}{2}(x - b)^2 + acx, \quad g_{\text{ex}}(x) = b - x + ac, \quad x^* = b + ac.$$

The reduced objective is 1-strongly concave and 1-smooth. For $0 < \eta_L \leq 1$ and compact intervals X, Y large enough that the displayed responses and stationary points are interior, write $\kappa_Q = 1 - \eta_L$. The common-base follower recursion has the closed form

$$y^{(K)}(x) = \kappa_Q^K y_{\text{init}} + ax(1 - \kappa_Q^K), \quad Z_K = \frac{\partial y^{(K)}}{\partial x} = a(1 - \kappa_Q^K).$$

Consequently,

$$F_K(x) = -\frac{1}{2}(x - b)^2 + c\{\kappa_Q^K y_{\text{init}} + ax(1 - \kappa_Q^K)\},$$

$$g_K(x) = b - x + ac(1 - \kappa_Q^K), \quad x_K^* = b + ac(1 - \kappa_Q^K),$$

and

$$|x_K^* - x^*| = |g_K(x) - g_{\text{ex}}(x)| = |ac|\kappa_Q^K.$$

The frozen-follower field is $g^{\text{dir}}(x) = b - x$, with zero $x^\dagger = b$. Thus $|x^\dagger - x^*| = |ac|$. Here $\mu_L = \mu_H = 1$, $M_{yx} = |a|$, and $\bar{g}_H = |c|$, so both inequalities in Eq. (16) are equalities. The finite-response bound is also exact: $\ell_{yy} = \ell_{yx} = 0$, hence $C_Z = 0$, $e_Z(K) = |a|\kappa_Q^K$, and Eq. (53) reduces to $\varepsilon_K = |ac|\kappa_Q^K$.

For the numerical illustration, take $a = 0.8$, $b = 0.1$, $c = 0.5$, $\eta_L = 0.25$, $y_{\text{init}} = 0$, and $\eta_H = 0.2$. Let $X = [-1, 1]$ and $Y = [-1.6, 1.6]$. Then $\xi(X) = [-0.8, 0.8]$, $D_0 = 0.8$, and the closed D_0 -tube around $\xi(X)$ is exactly Y . Both x^* and the direct-only stationary point $x^\dagger$ are interior, and the displayed outer trajectories from $x_0 = -0.4$ remain in X . Over $K \in \{1, 2, 4, 8, 16\}$, centered finite differences of F_K agree with g_K to a maximum absolute error of 9.6×10^{-12} , while the computed target errors agree with $|ac|\kappa_Q^K$ to 5.6×10^{-17} .

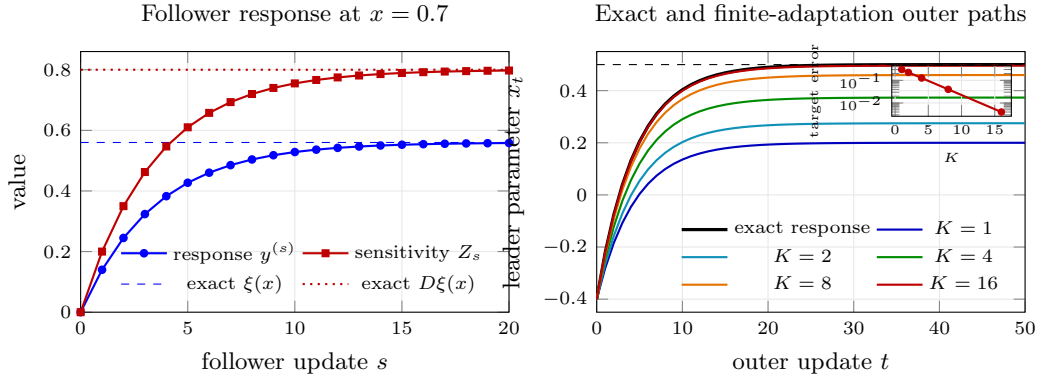


Figure 2: Closed-form and numerical checks for the quadratic family. Left: the finite response and sensitivity at $x = 0.7$ approach $\xi(x) = 0.56$ and $D\xi(x) = 0.8$. Center: exact response-aware ascent and finite- K ascent, with $x^* = 0.5$ dashed. Inset: target error and its exact value $|ac|\kappa_Q^K$.

G.2 BERNOULLI POLICY-COORDINATE WITNESS

Let $x_{\max}, \alpha, \gamma > 0$, $0 < \epsilon < 1/2$, and $\omega_B, v_B \neq 0$. Take

$$X = [-x_{\max}, x_{\max}], \quad Y = [\epsilon, 1 - \epsilon],$$

and let the follower coordinate y be the probability of one action:

$$\begin{aligned} u_L(x, y) &= \omega_B x y + \alpha [-y \log y - (1 - y) \log(1 - y)], \\ u_H(x, y) &= -\frac{\gamma}{2} x^2 + v_B y. \end{aligned}$$

Writing $\sigma(z) = (1 + e^{-z})^{-1}$, follower stationarity gives

$$\xi(x) = \sigma(\omega_B x / \alpha), \quad D\xi(x) = \frac{\omega_B}{\alpha} \xi(x)(1 - \xi(x)).$$

If $\epsilon < \sigma(-|\omega_B| x_{\max} / \alpha)$, the response is in $\text{int}(Y)$. Since

$$\nabla_{yy}^2 u_L(x, y) = -\frac{\alpha}{y(1 - y)}, \quad \nabla_{yx}^2 u_L(x, y) = \omega_B,$$

valid follower constants are

$$\mu_L = 4\alpha, \quad L_L = \frac{\alpha}{\epsilon(1 - \epsilon)}, \quad M_{yx} = |\omega_B|.$$

The reduced objective is

$$F(x) = -\frac{\gamma}{2} x^2 + v_B \sigma(\omega_B x / \alpha).$$

The elementary bound

$$|\xi''(x)| \leq \frac{\omega_B^2}{6\sqrt{3}\alpha^2}$$

follows by maximizing $|p(1 - p)(1 - 2p)|$ over $p \in [0, 1]$. Hence the conditions

$$\gamma > \frac{|v_B| \omega_B^2}{6\sqrt{3}\alpha^2}, \quad \gamma x_{\max} > \frac{|v_B \omega_B|}{4\alpha}$$

make F strongly concave and force its derivative to point inward at both endpoints. Therefore $x^* \in \text{int}(X)$, with certified constants

$$\mu_H = \gamma - \frac{|v_B| \omega_B^2}{6\sqrt{3}\alpha^2}, \quad L_F \leq \gamma + \frac{|v_B| \omega_B^2}{6\sqrt{3}\alpha^2}.$$

The direct field is $g^{\text{dir}}(x) = -\gamma x$, so $x^\dagger = 0$. Proposition 1 gives

$$|x^*| \leq \frac{|\omega_B v_B|}{4\alpha \mu_H}.$$

A concrete choice is

$$(\alpha, \omega_B, v_B, \gamma, x_{\max}, \epsilon, y_{\text{init}}) = (1, 1, 1, 1, 0.5, 0.1, 0.5).$$

It gives $\xi(X) = [0.37754, 0.62246]$ and $D_0 = 0.12246$. The closed tube is $[0.25508, 0.74492] \subset Y = [0.1, 0.9]$. Moreover, $\mu_H \approx 0.9037750$, $L_F < 1.0962251$, and the unique optimum is $x^* = 0.2462481$. The choices $0 < \eta_L \leq 1/L_L = 0.09$ and $\eta_H = 0.5$ satisfy both step-size conditions. All remaining derivatives used by the finite-response result are bounded throughout an open neighborhood containing both compact intervals.

H USE OF LANGUAGE MODELS

Language-model assistance supported literature discovery, organization, drafting, notation and consistency checks, proof review, counterexample analysis, and validation-test design. The authors independently inspected the resulting arguments and retain responsibility for the mathematics, empirical claims, citation accuracy, and final manuscript.